

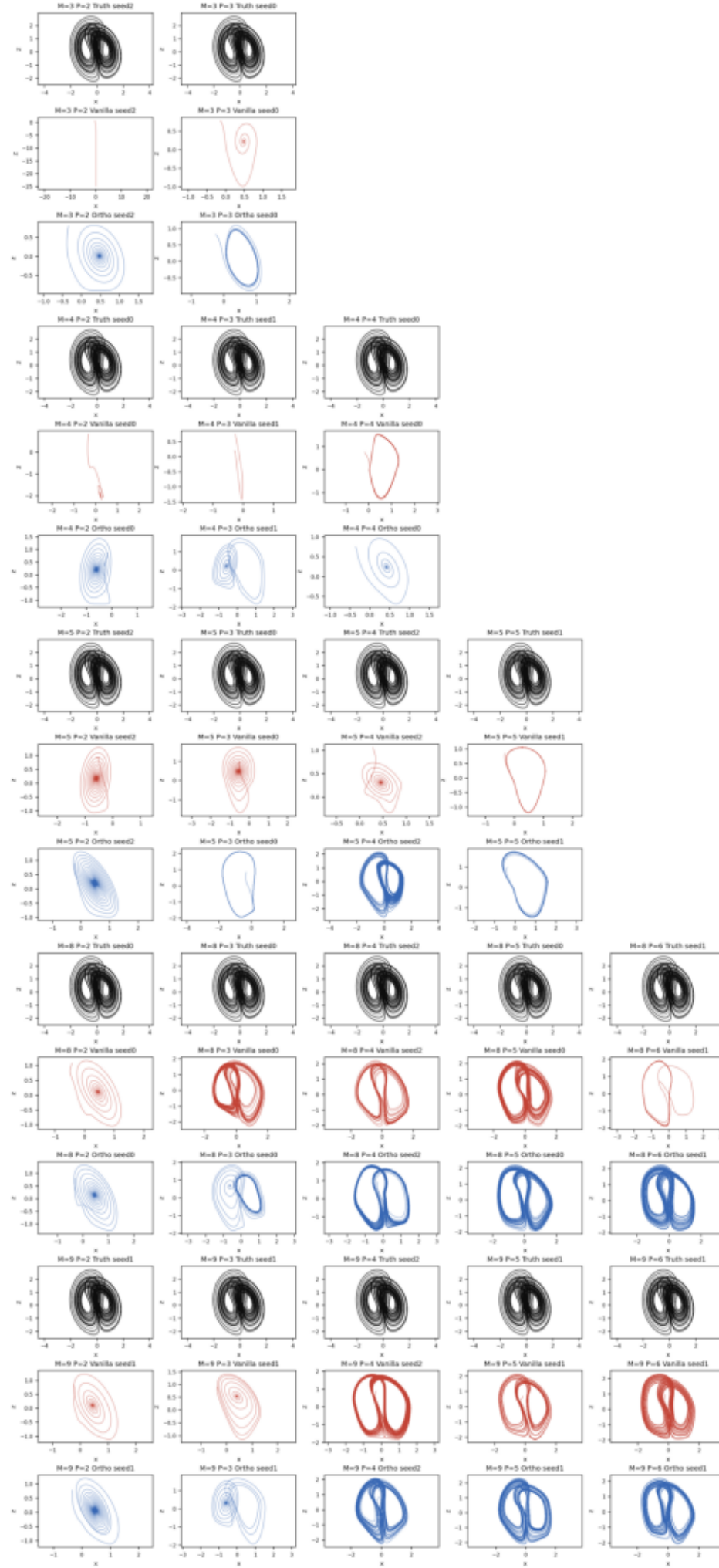
Seedrun v2 visuals: trajectories + 3-D (M, P, metric)

This file collects the trajectory comparison and the (M, P, metric) visualisations for the seedrun_v2 sweep.

Sections:

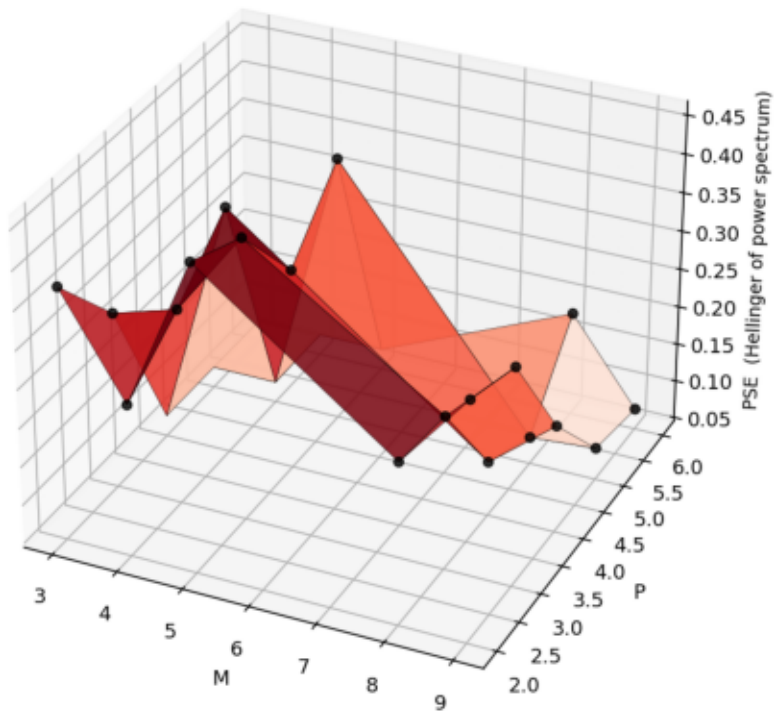
1. Wall image: every (M, P) cell at the median-PSE seed.
2. 3-D performance surfaces (M x P x metric) for both models.
3. 3-D bar plots (M x P x metric) and gap surfaces.
4. M-cross-sections: metric vs M, one curve per P.
5. P-cross-sections: metric vs P, one curve per M.
6. Per-cell trajectories at high resolution: 19 pages showing all 3 seeds in 3-D and x-z 2-D.

Trajectory wall (v2): all (M, P) cells, median-PSE seed, x-z projection

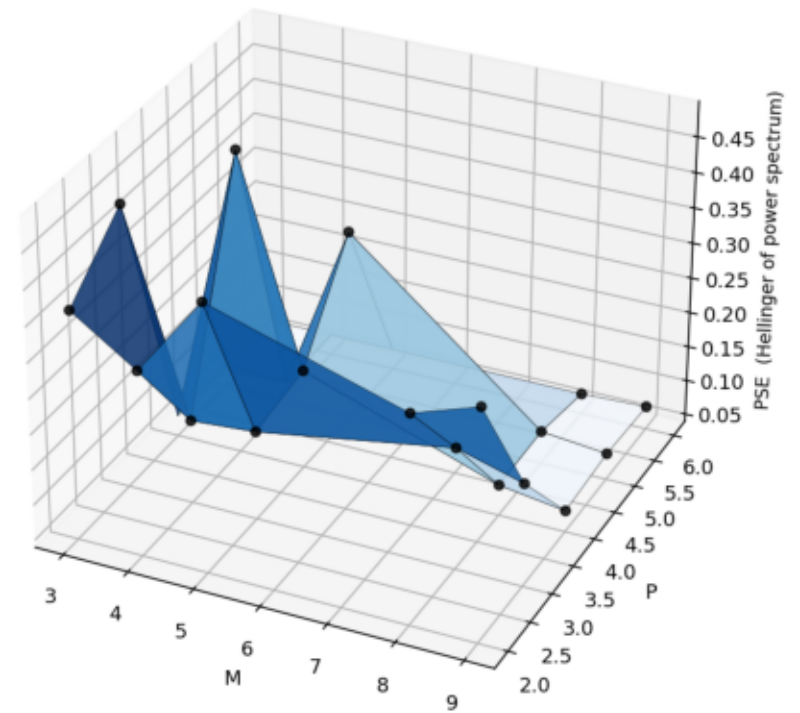


3-D performance surface: PSE (Hellinger of power spectrum)

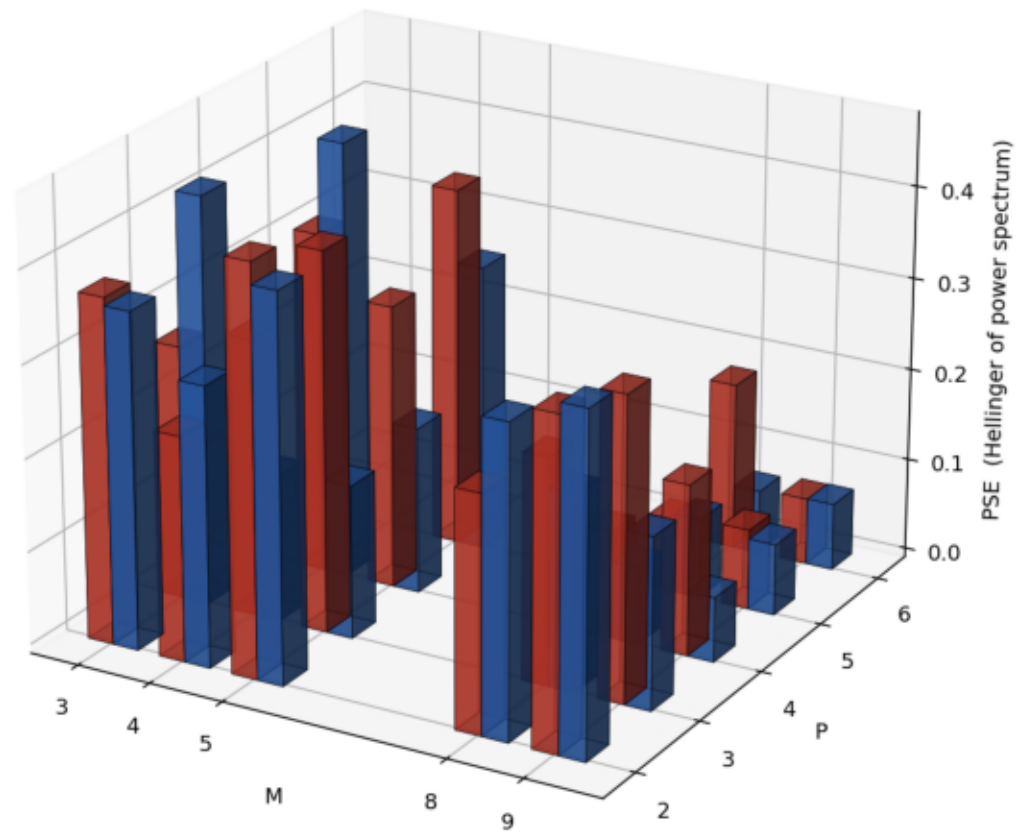
Vanilla: PSE (Hellinger of power spectrum)



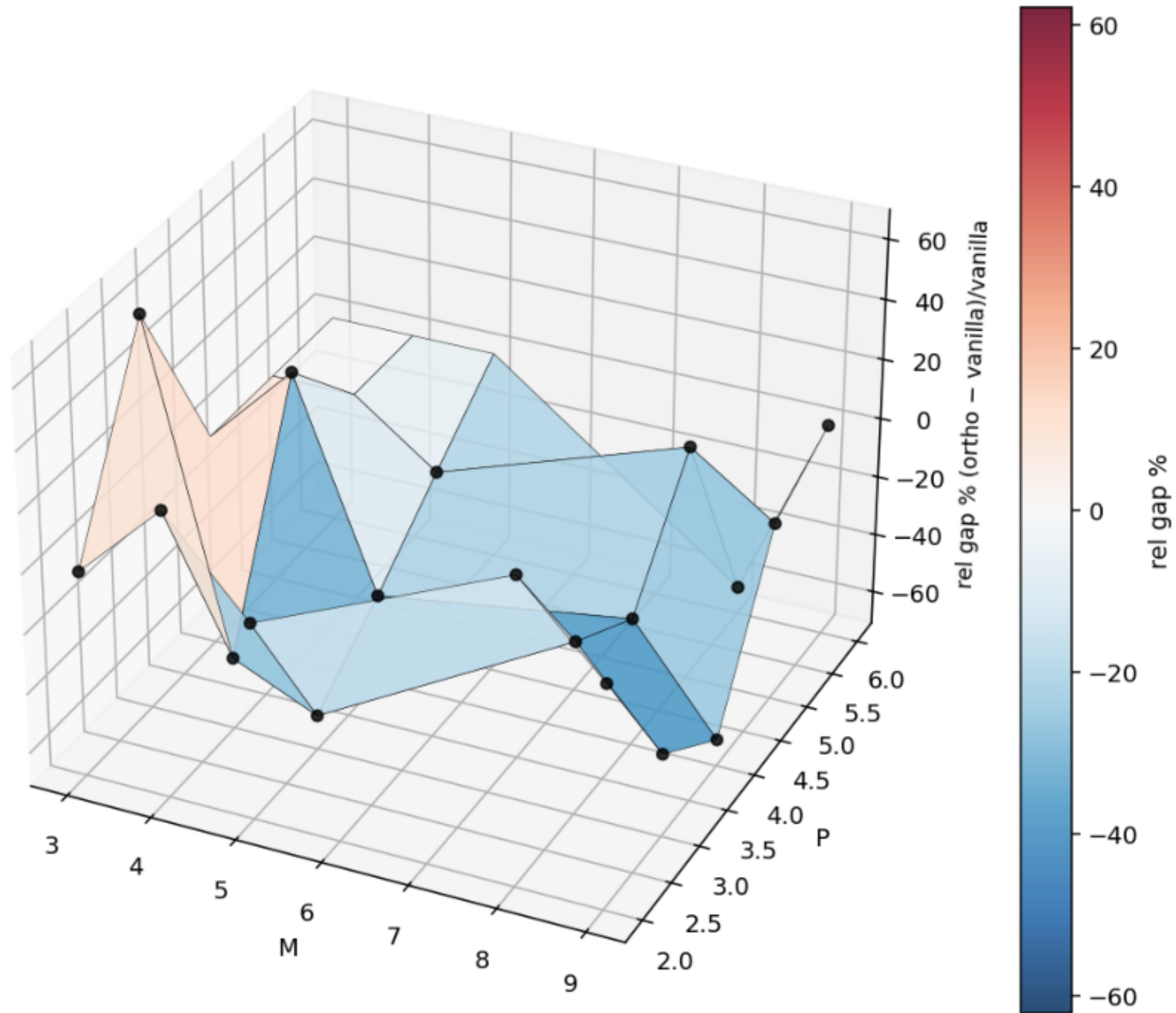
Orthogonal: PSE (Hellinger of power spectrum)



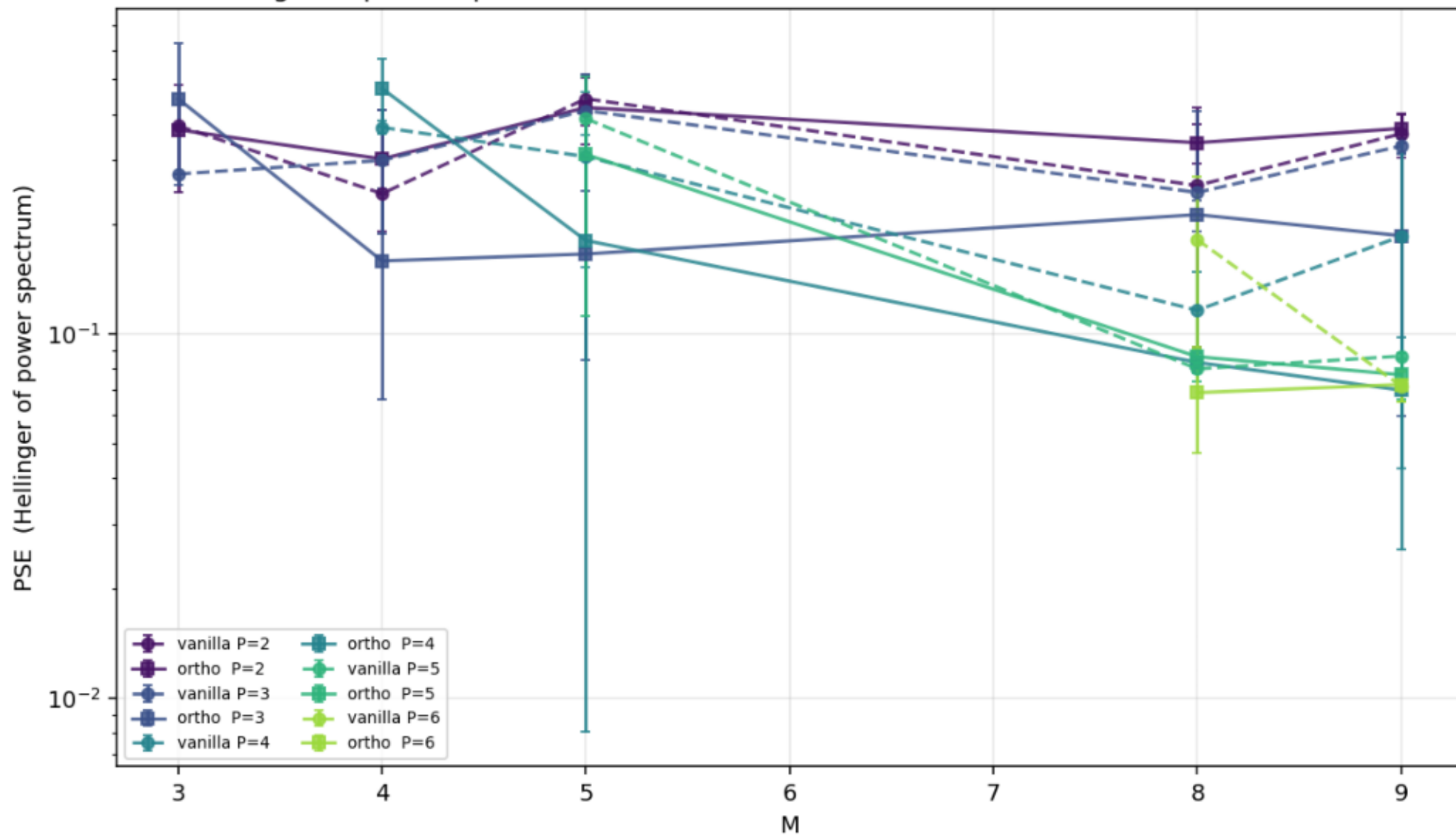
3-D bars: PSE (Hellinger of power spectrum) (red = vanilla, blue = ortho)



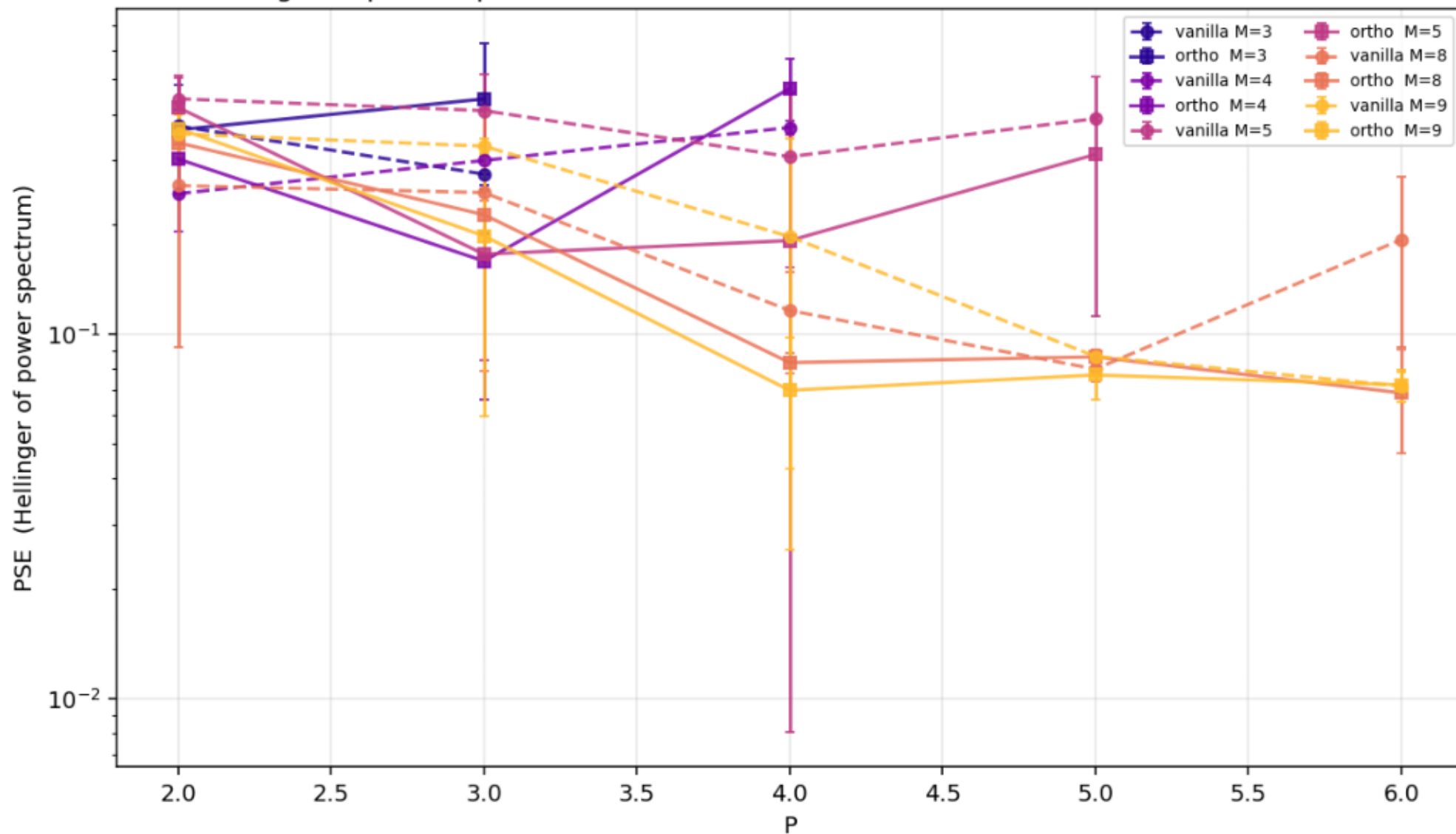
3-D gap surface: PSE (Hellinger of power spectrum)
blue = ortho wins, red = vanilla wins



PSE (Hellinger of power spectrum) vs M (vanilla dashed, ortho solid; mean $\pm$ std over 3 seeds)

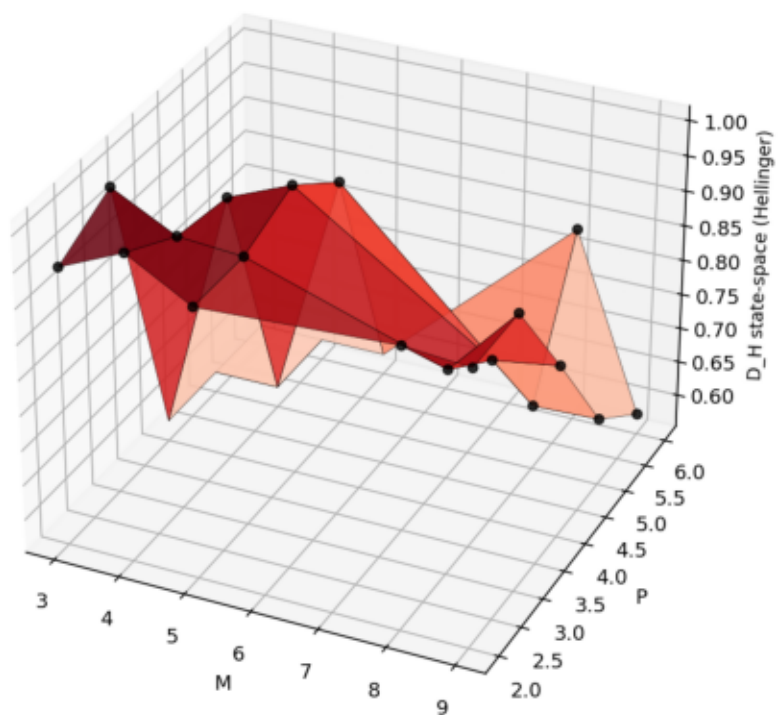


PSE (Hellinger of power spectrum) vs P (vanilla dashed, ortho solid; mean $\pm$ std over 3 seeds)

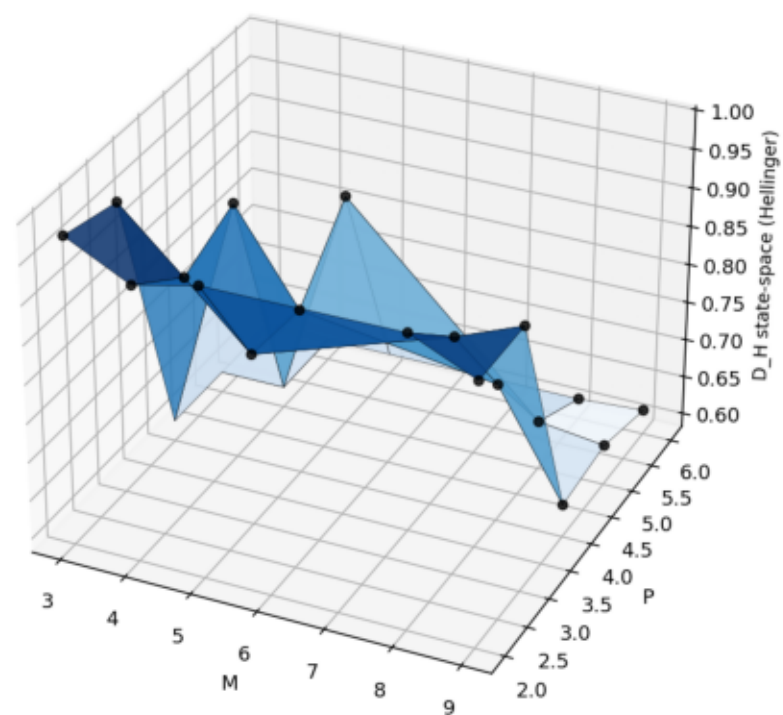


3-D performance surface: D_H state-space (Hellinger)

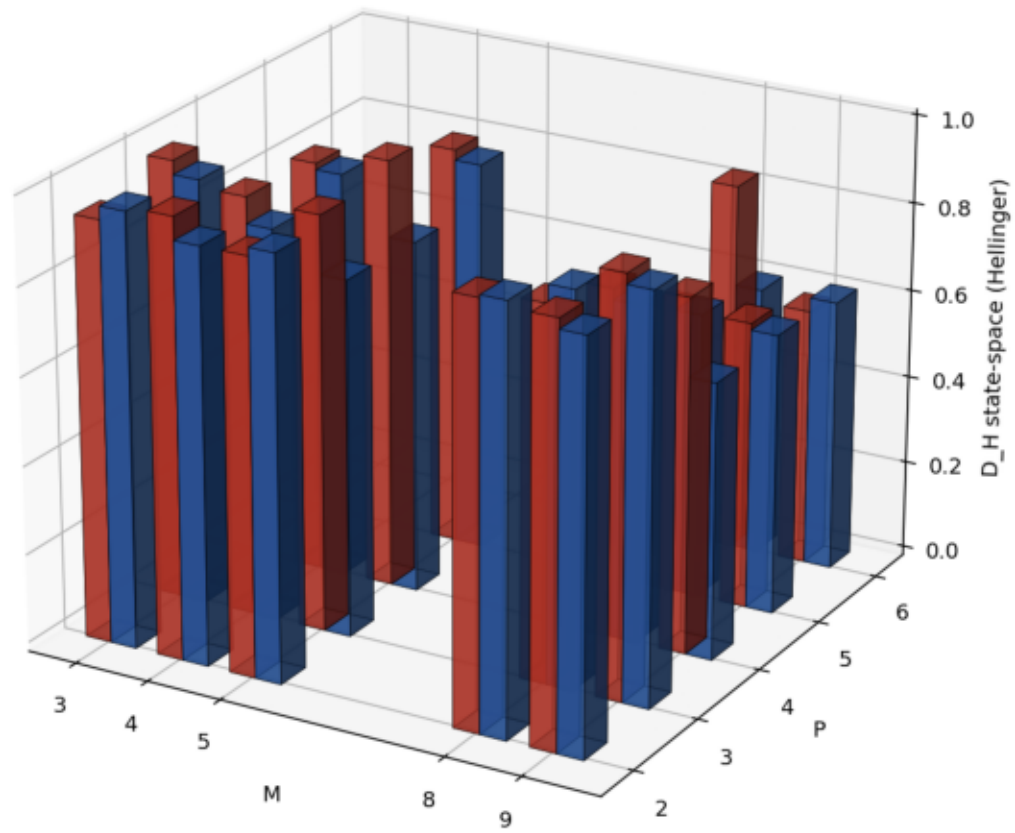
Vanilla: D_H state-space (Hellinger)



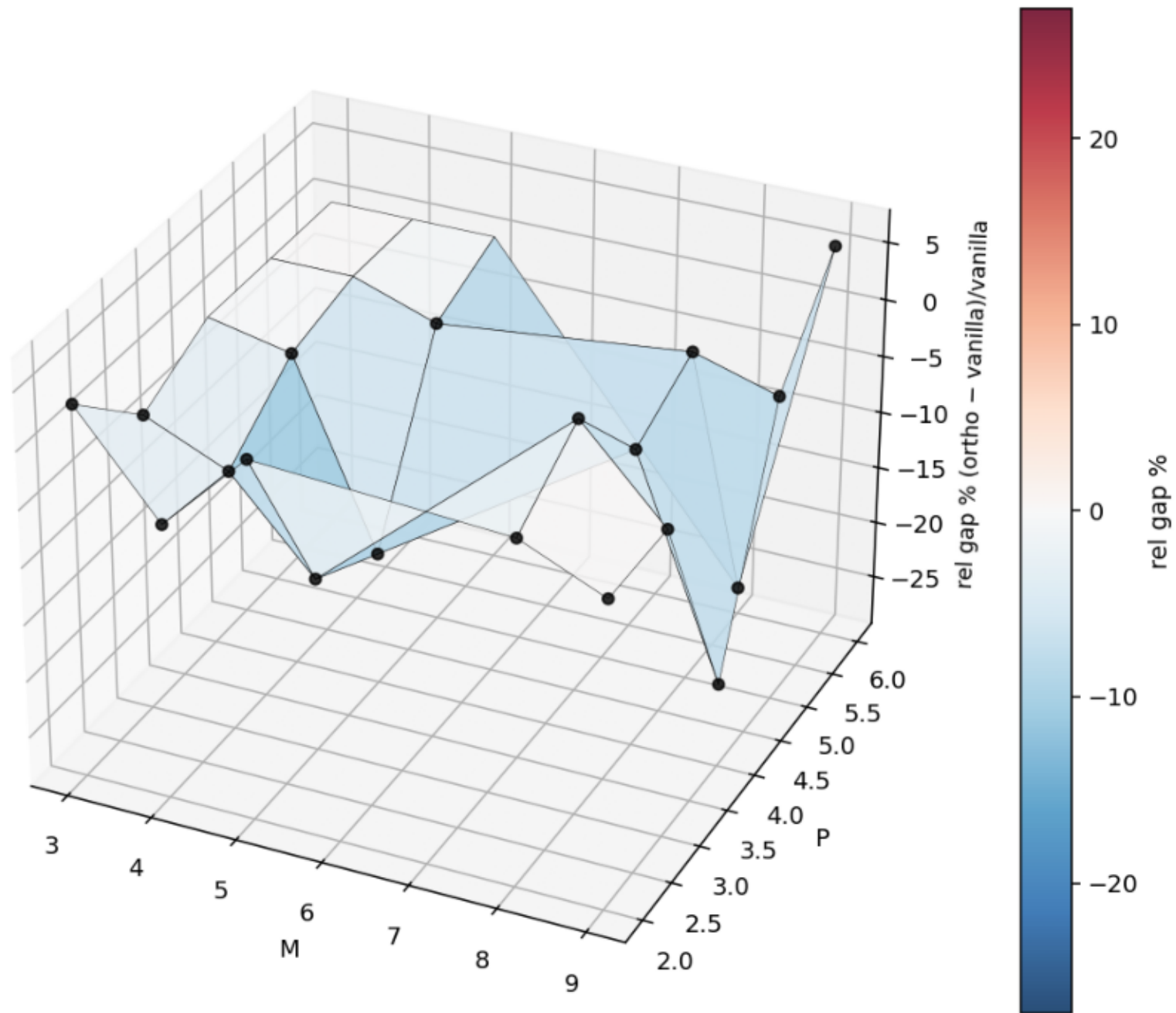
Orthogonal: D_H state-space (Hellinger)



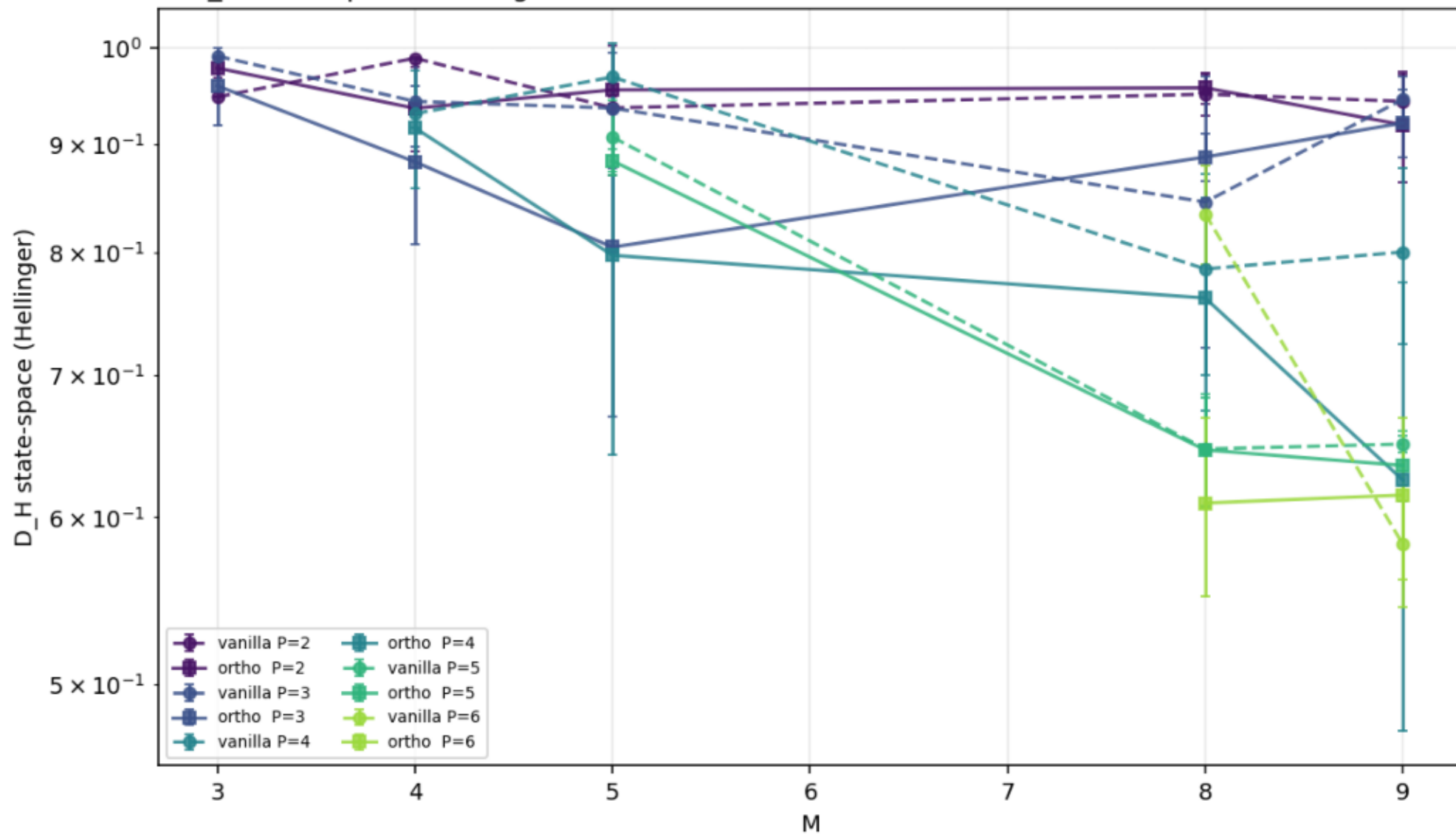
3-D bars: D_H state-space (Hellinger) (red = vanilla, blue = ortho)



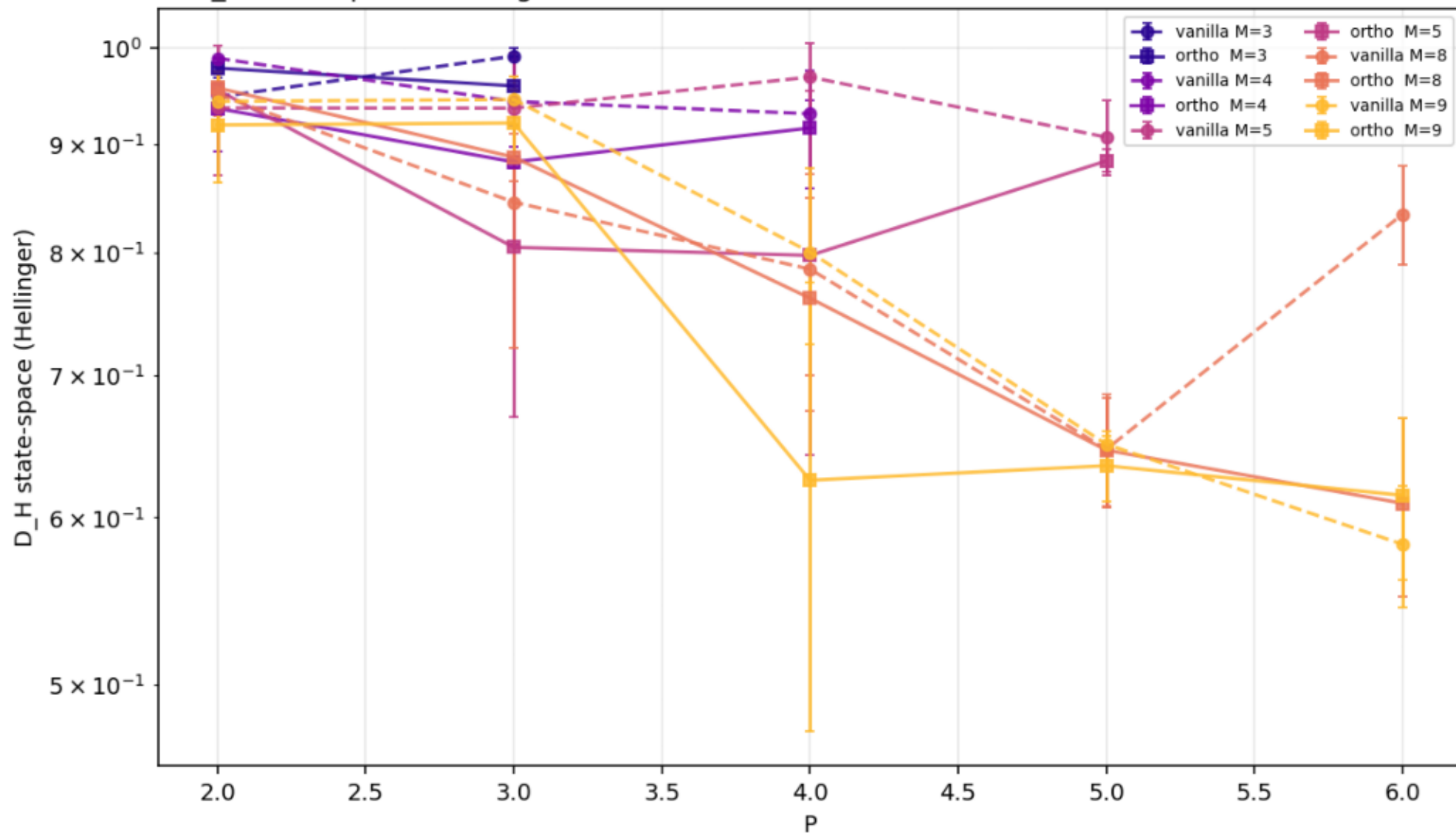
3-D gap surface: D_H state-space (Hellinger)
blue = ortho wins, red = vanilla wins



D_H state-space (Hellinger) vs M (vanilla dashed, ortho solid; mean $\pm$ std over 3 seeds)

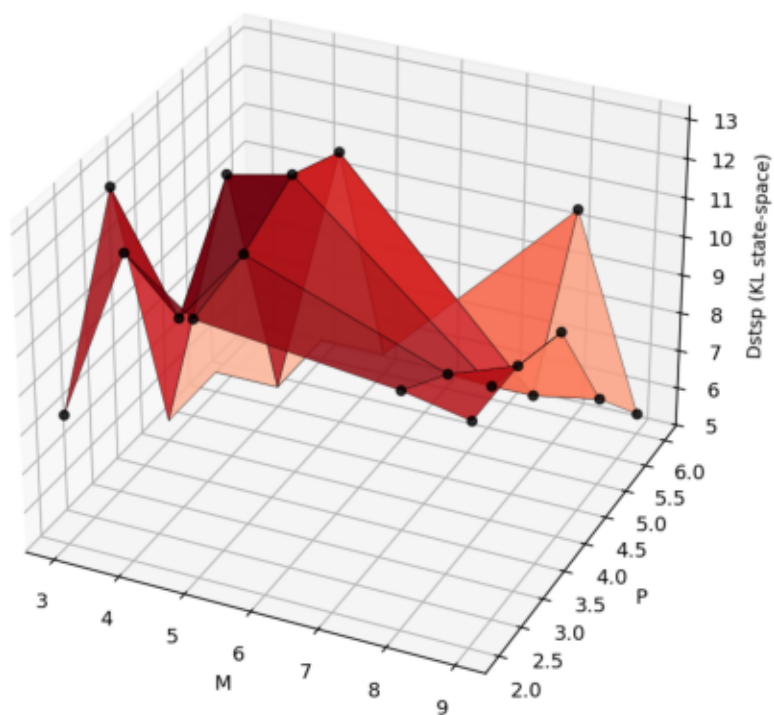


D_H state-space (Hellinger) vs P (vanilla dashed, ortho solid; mean +/- std over 3 seeds)

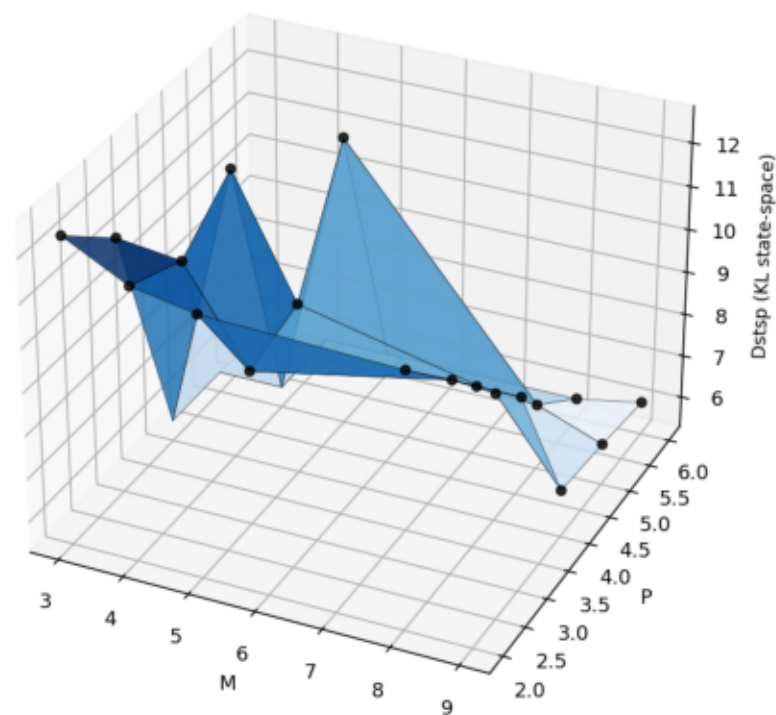


3-D performance surface: Dstsp (KL state-space)

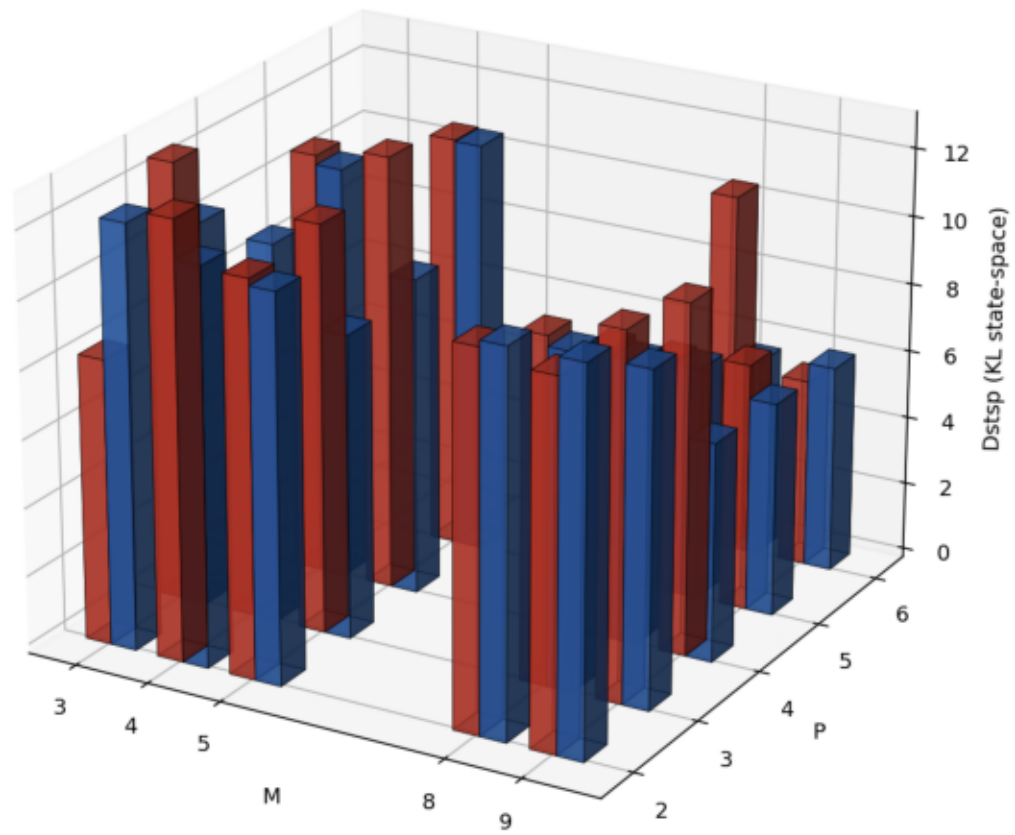
Vanilla: Dstsp (KL state-space)



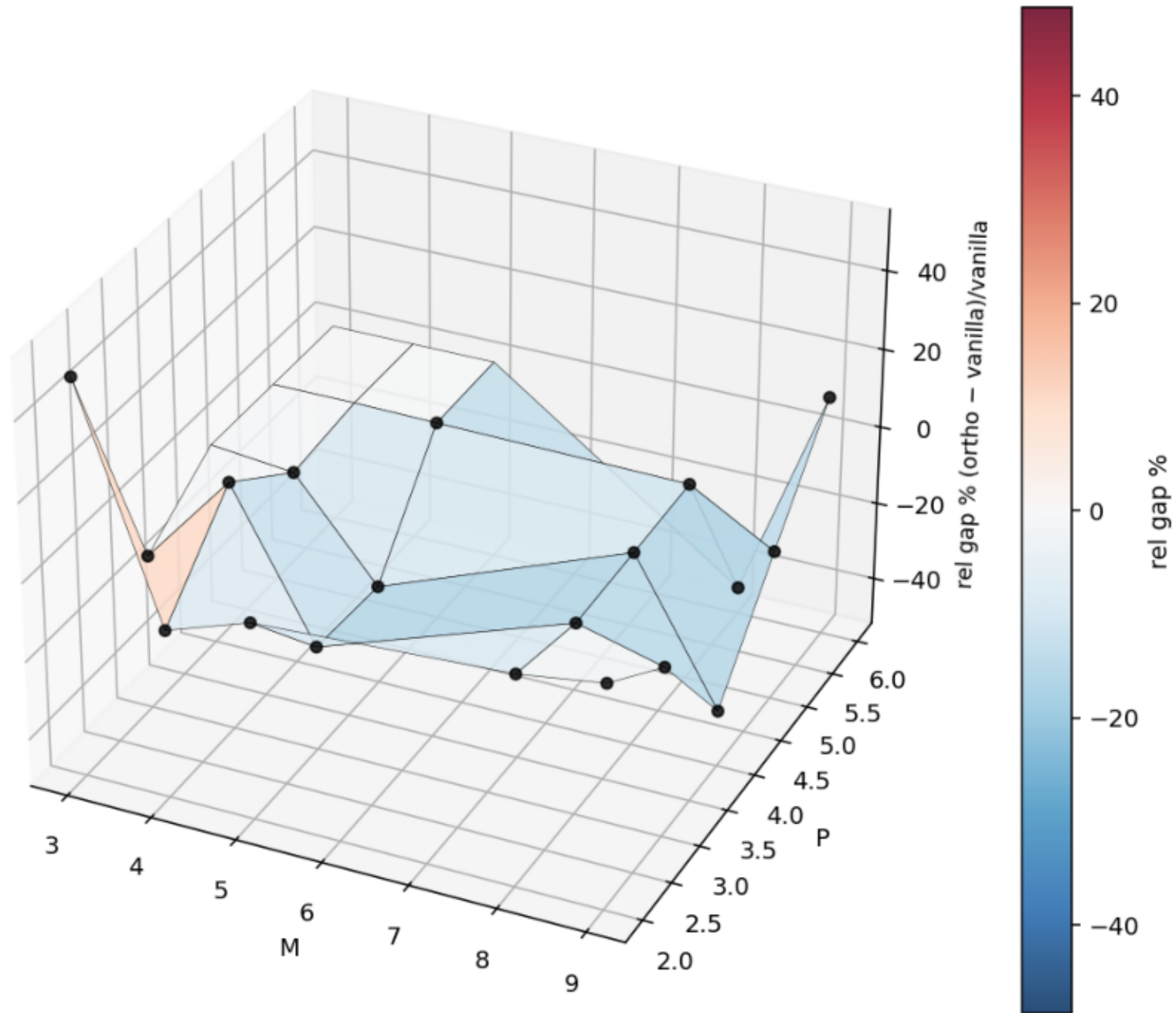
Orthogonal: Dstsp (KL state-space)



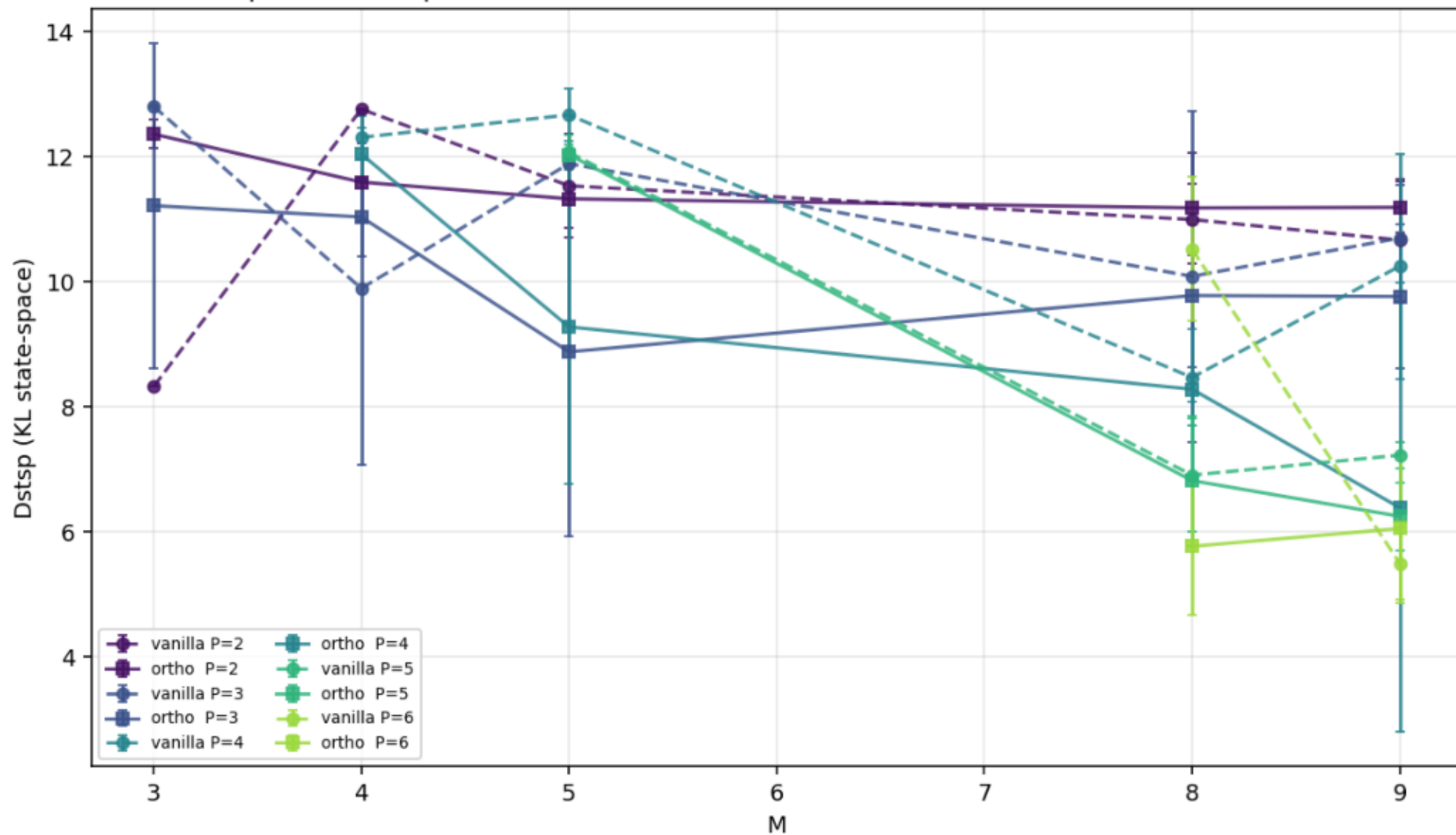
3-D bars: Dstsp (KL state-space) (red = vanilla, blue = ortho)



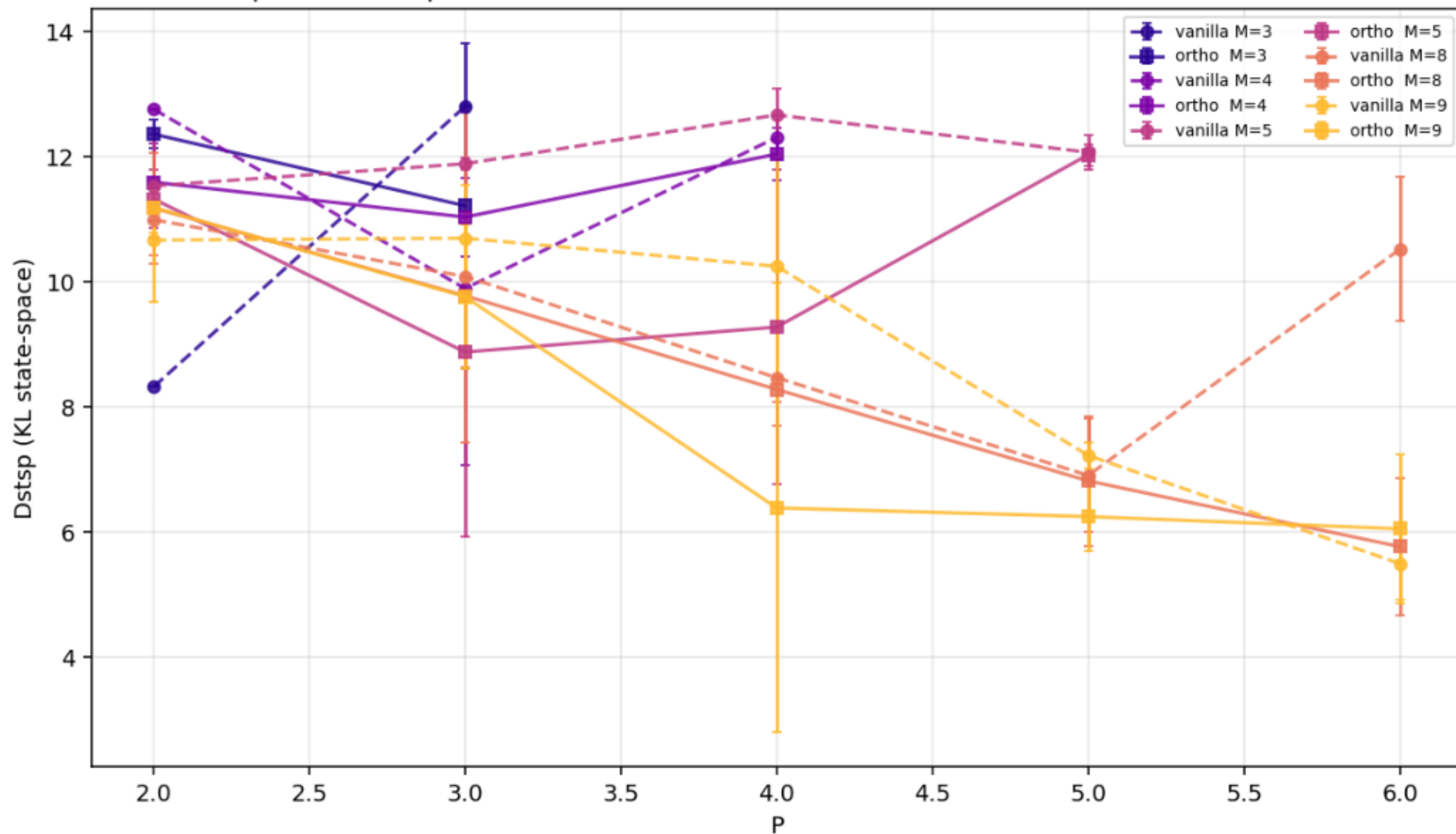
3-D gap surface: Dstsp (KL state-space)
blue = ortho wins, red = vanilla wins



Dstsp (KL state-space) vs M (vanilla dashed, ortho solid; mean $\pm$ std over 3 seeds)

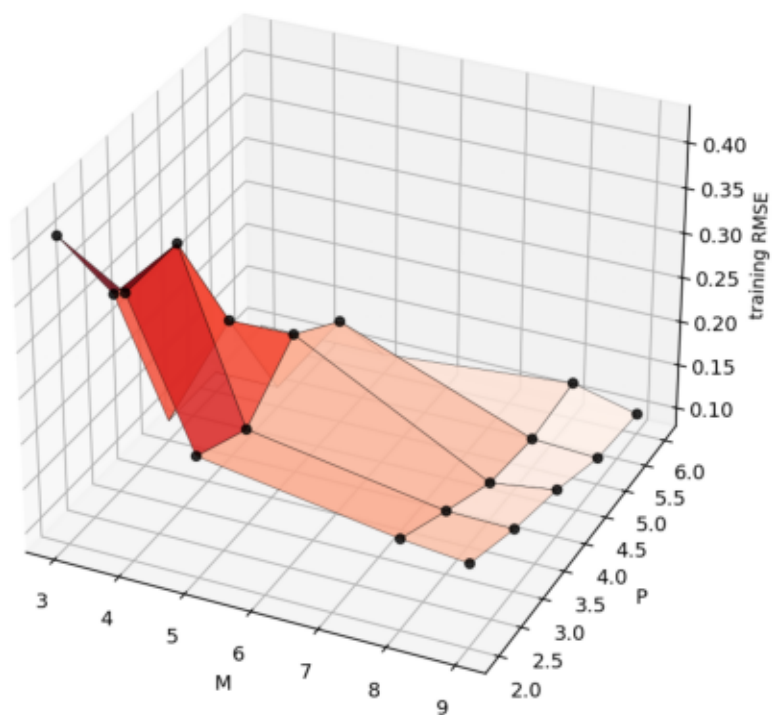


Dstsp (KL state-space) vs P (vanilla dashed, ortho solid; mean $\pm$ std over 3 seeds)

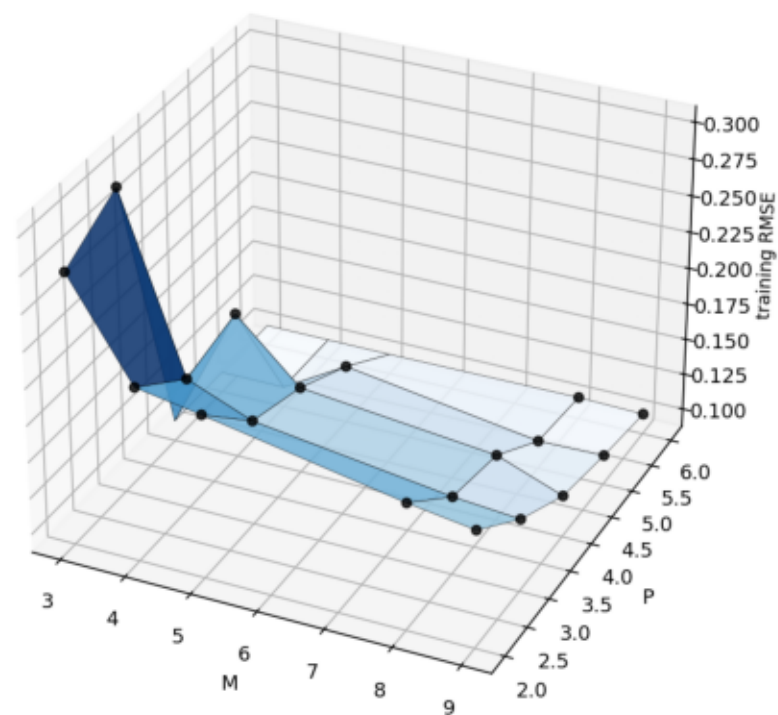


3-D performance surface: training RMSE

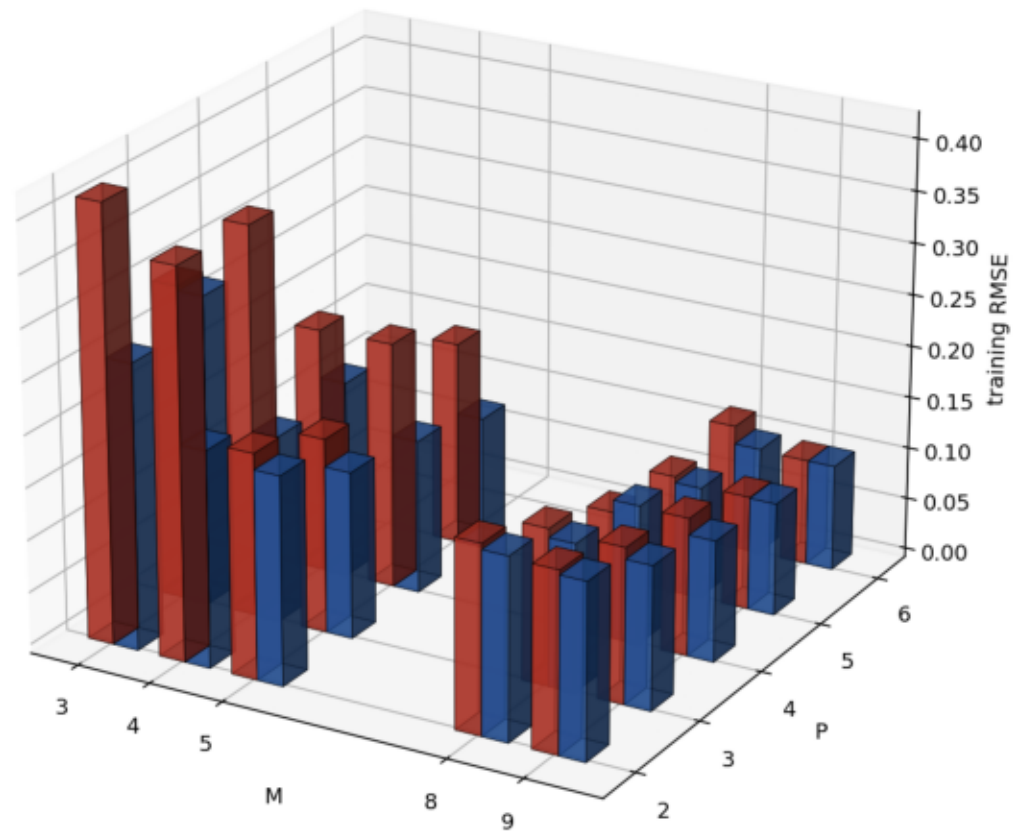
Vanilla: training RMSE



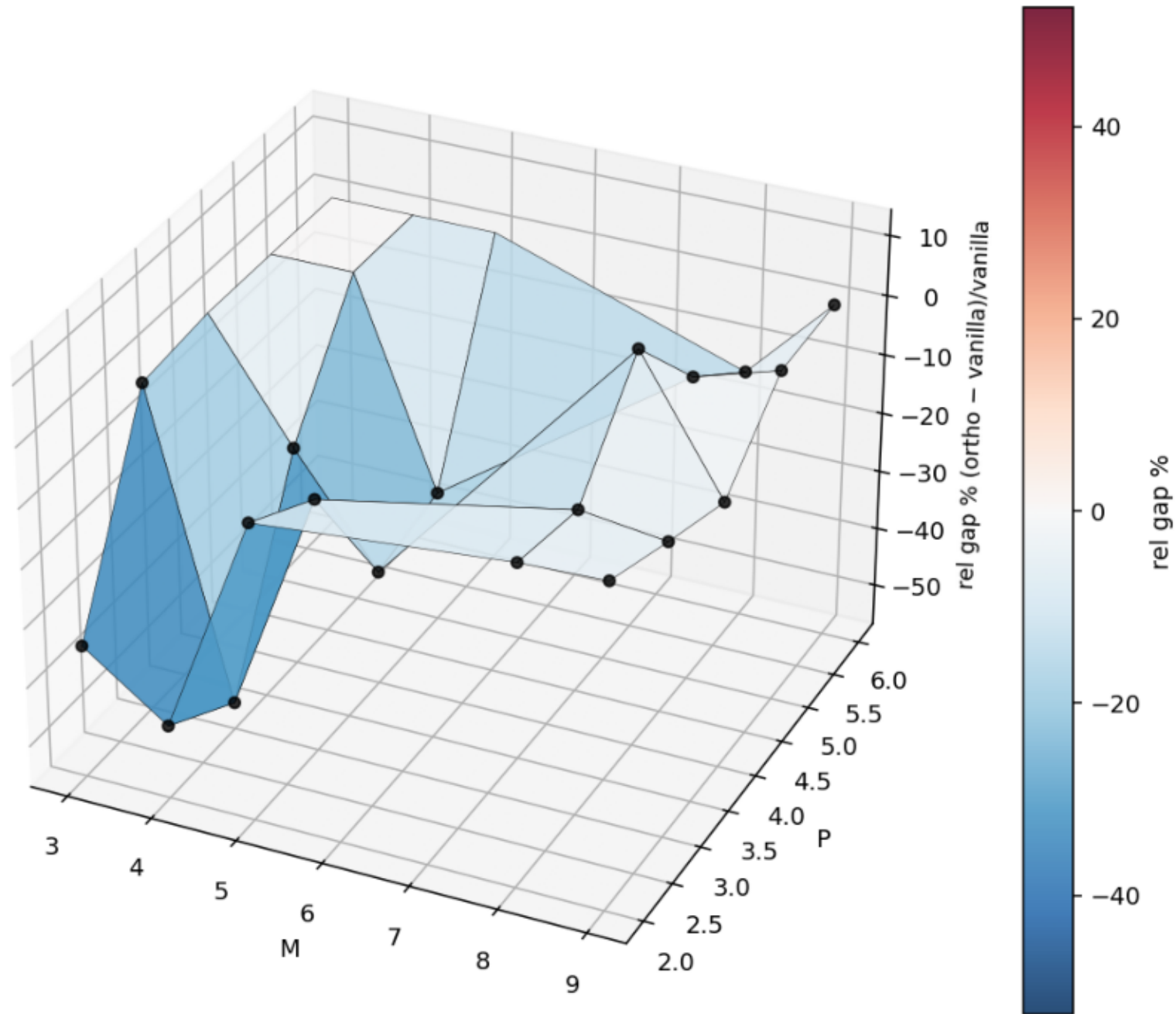
Orthogonal: training RMSE



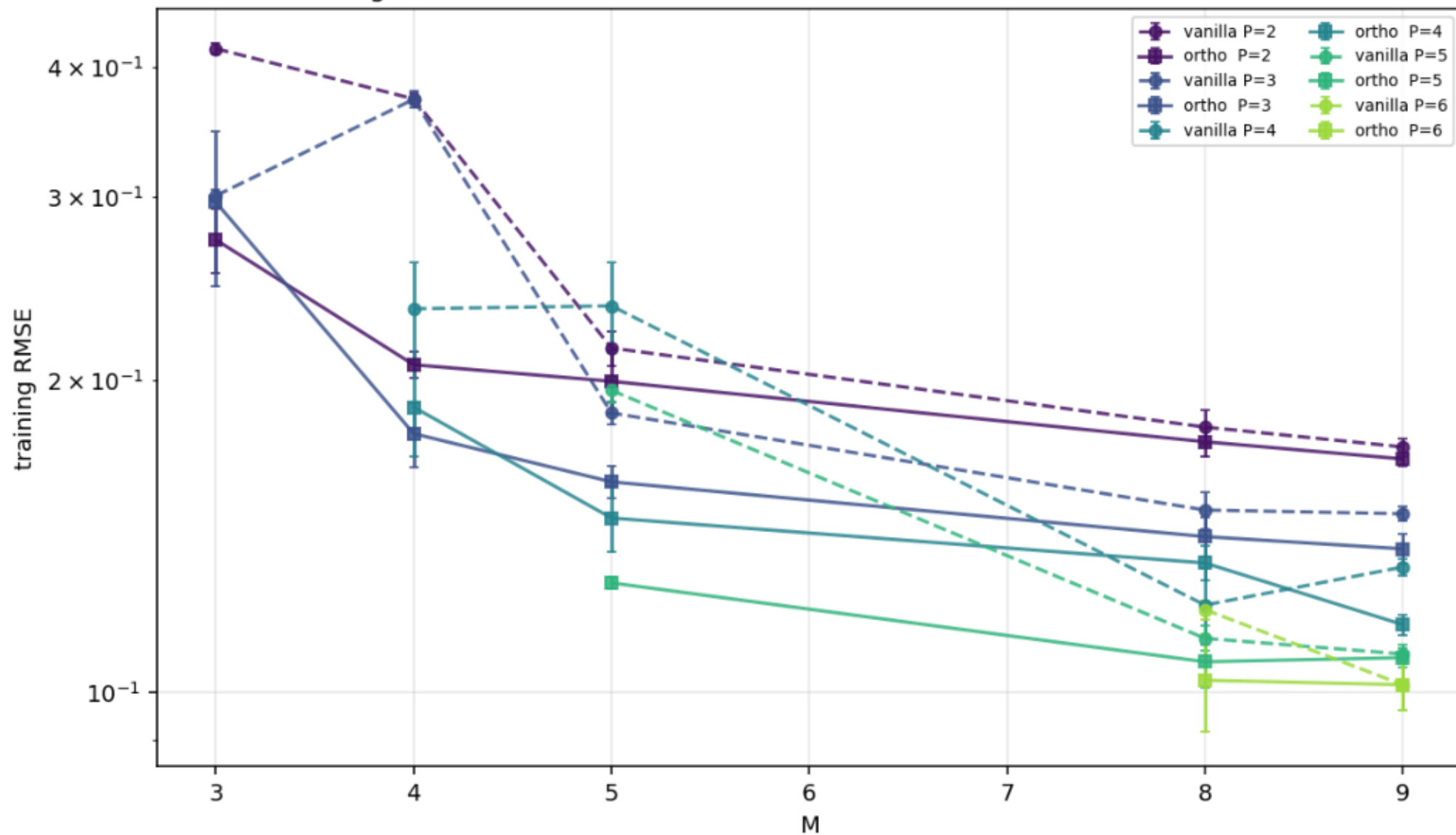
3-D bars: training RMSE (red = vanilla, blue = ortho)



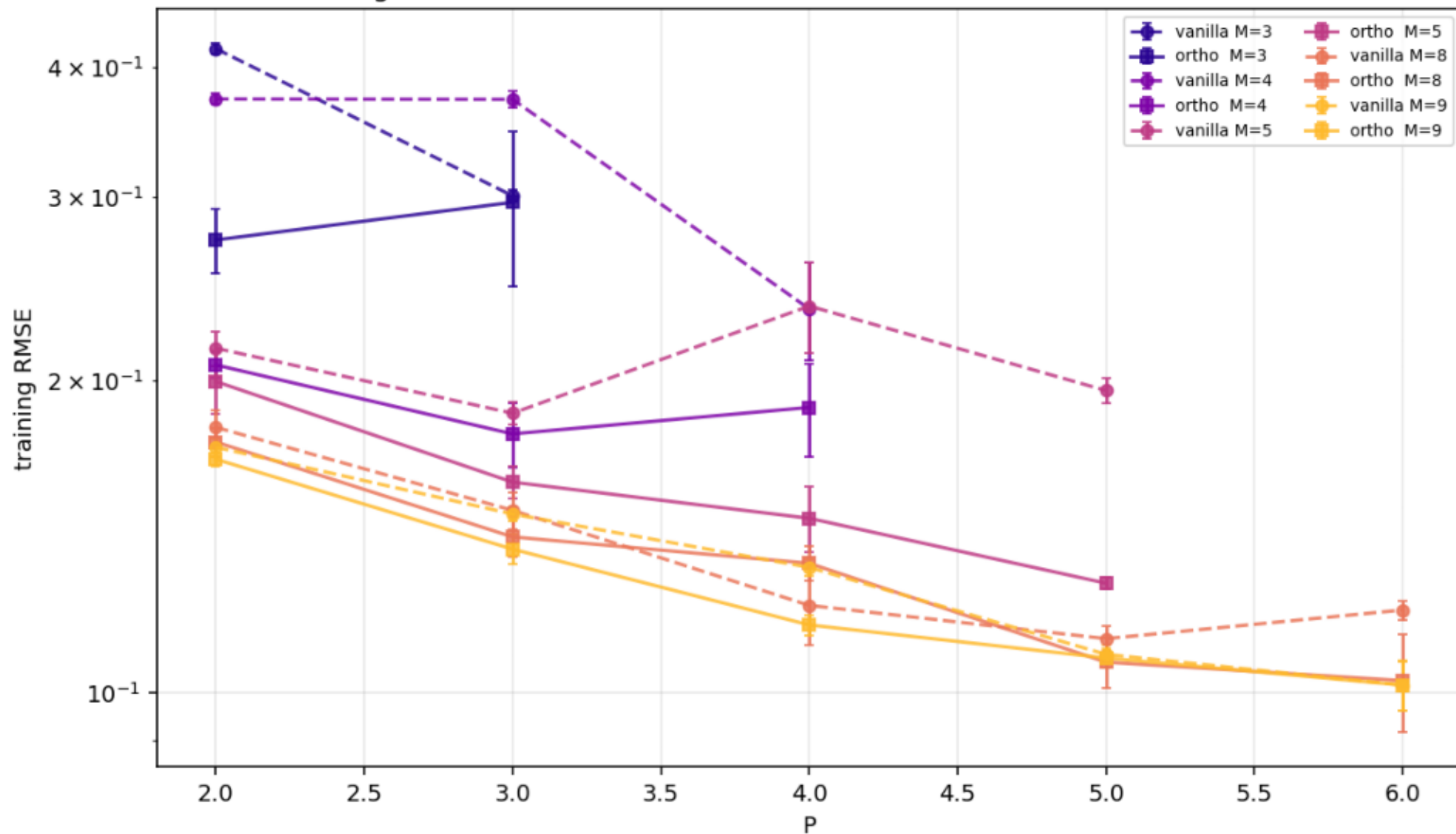
3-D gap surface: training RMSE
blue = ortho wins, red = vanilla wins



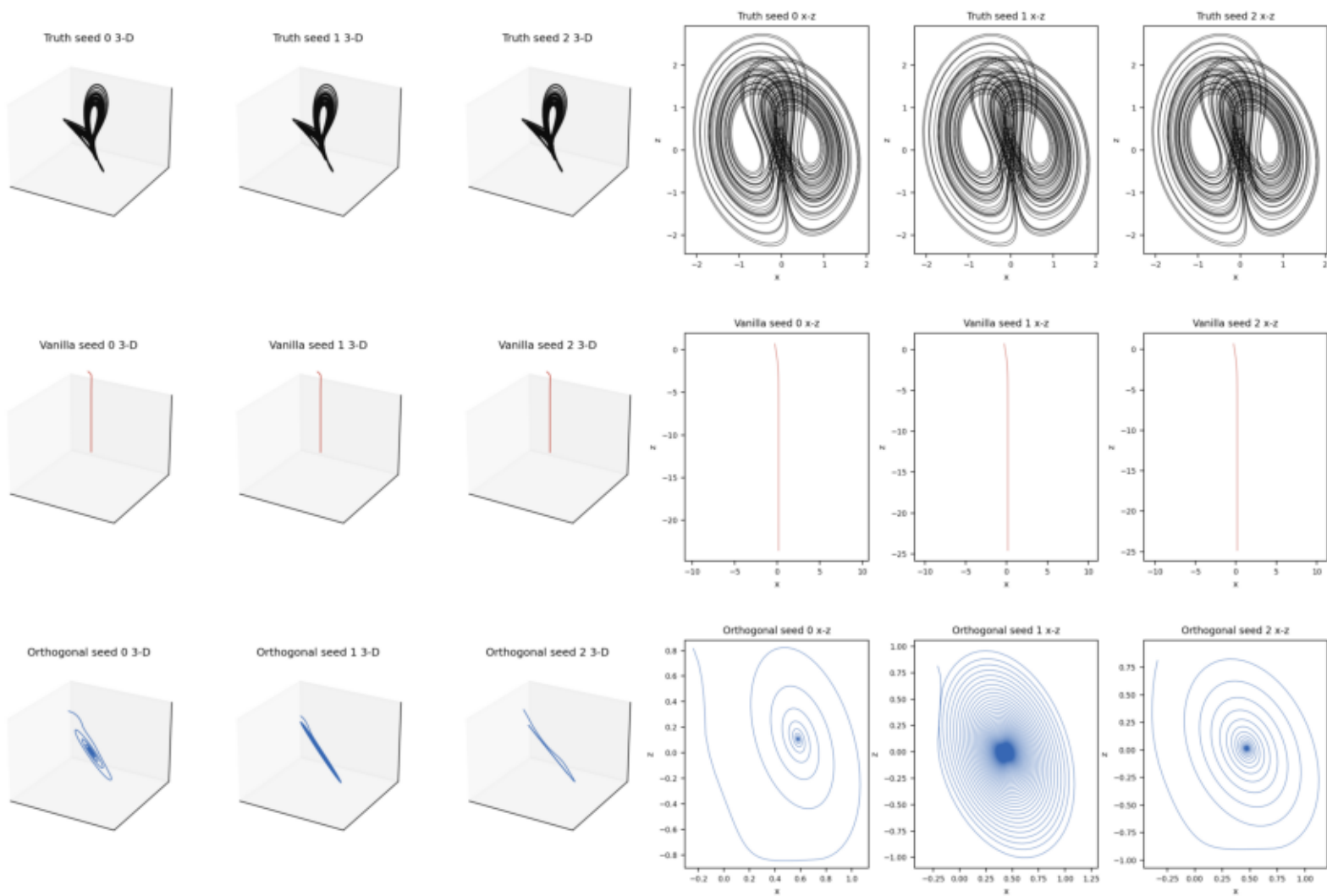
training RMSE vs M (vanilla dashed, ortho solid; mean +/- std over 3 seeds)



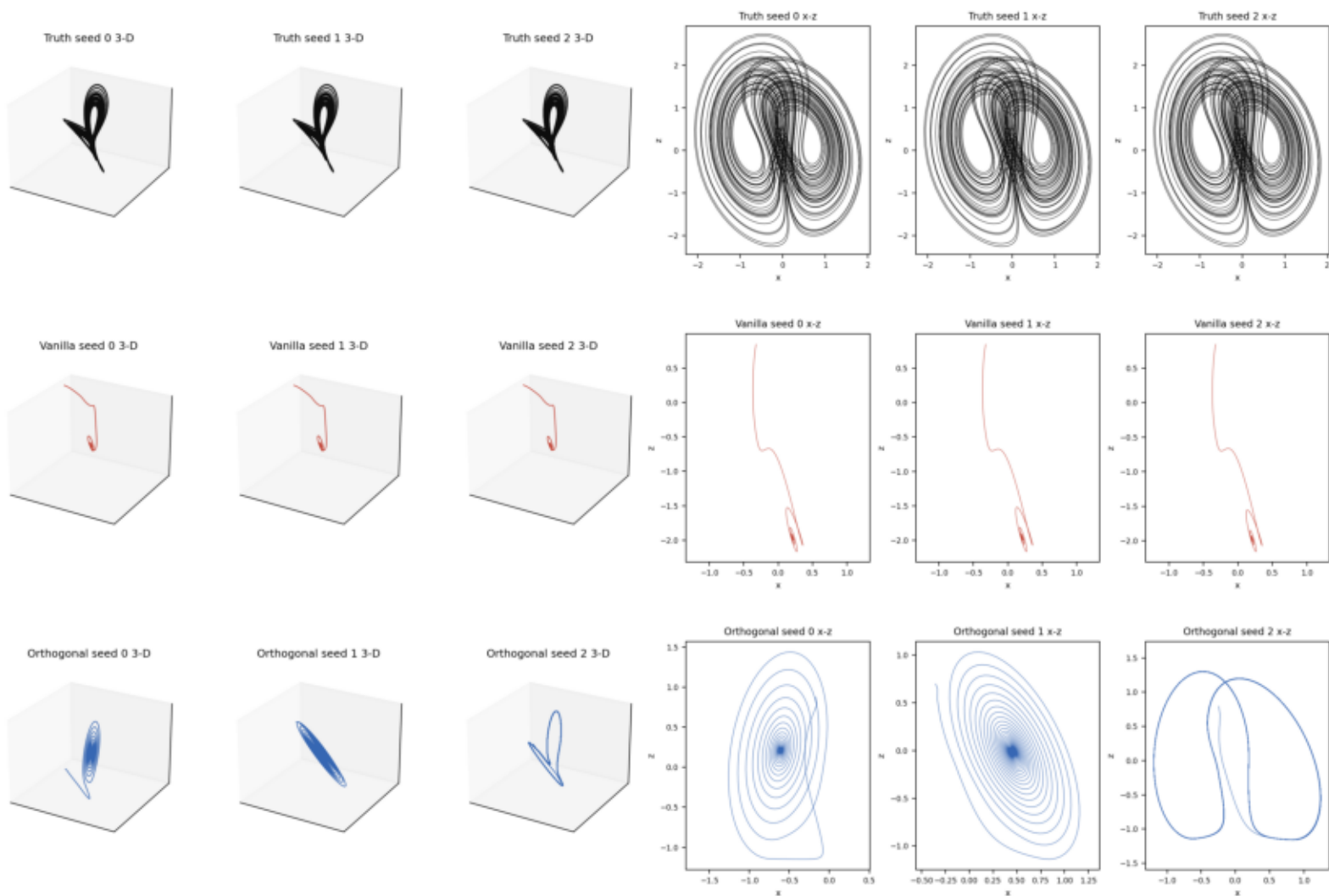
training RMSE vs P (vanilla dashed, ortho solid; mean +/- std over 3 seeds)



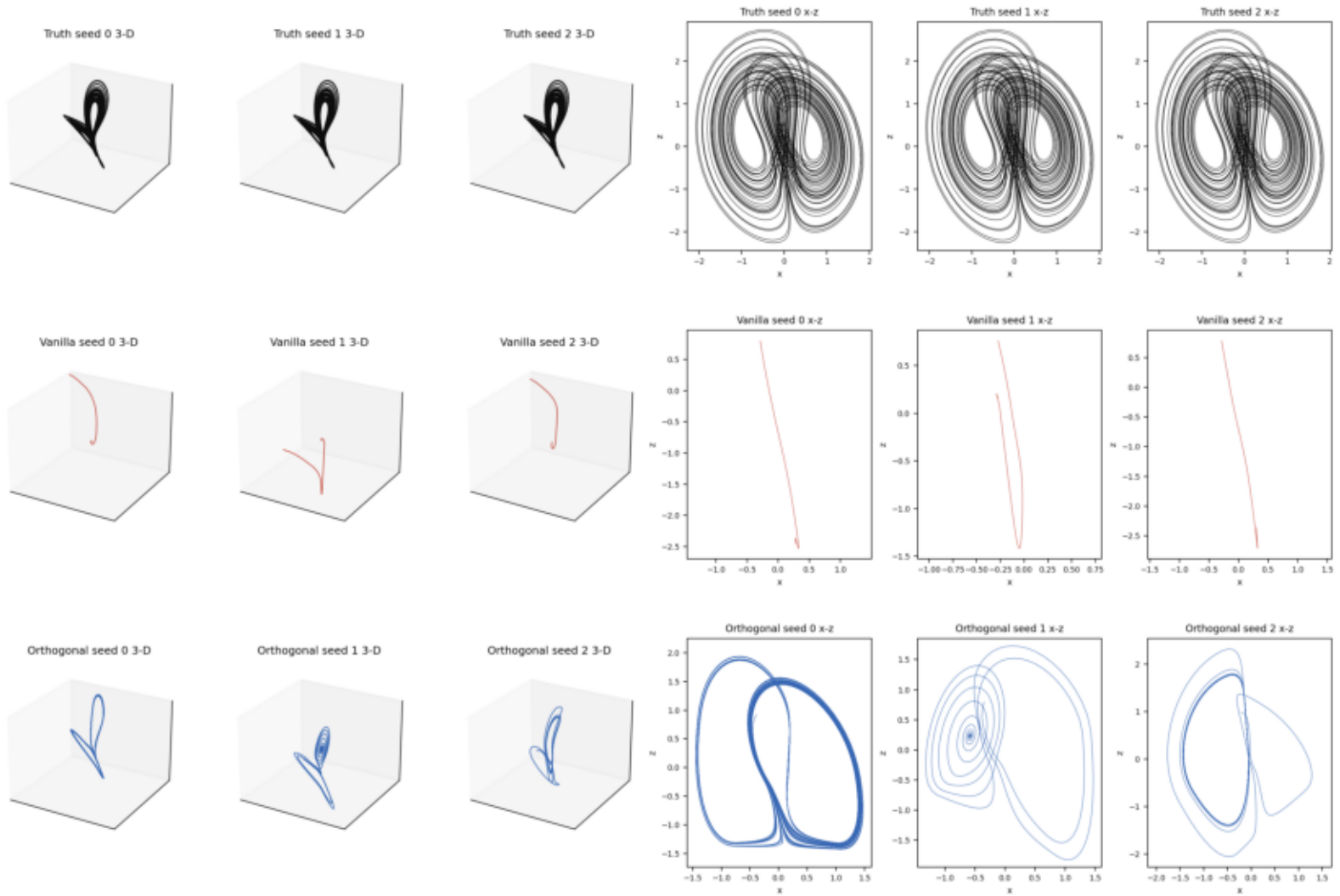
M = 3 P = 2 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3712+/-0.0007 ortho 0.3624+/-0.1180 rel gap -2.4% D_H_state van 0.948+/-0.000 ortho 0.978+/-0.010 rel gap +3.2%



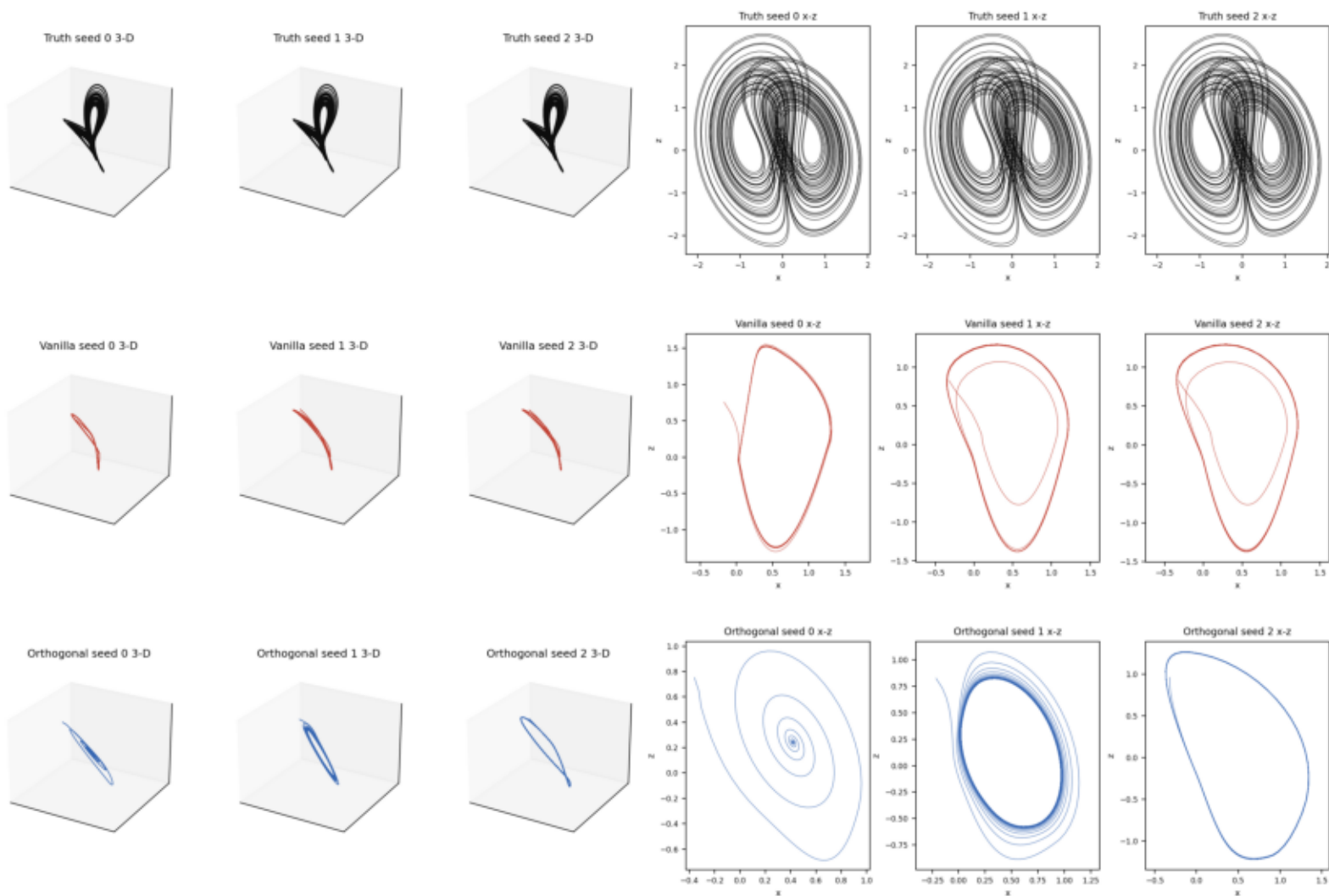
M = 4 P = 2 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.2421+/-0.0023 ortho 0.3014+/-0.1106 rel gap +24.5% D_H_state van 0.988+/-0.000 ortho 0.936+/-0.042 rel gap -5.3%



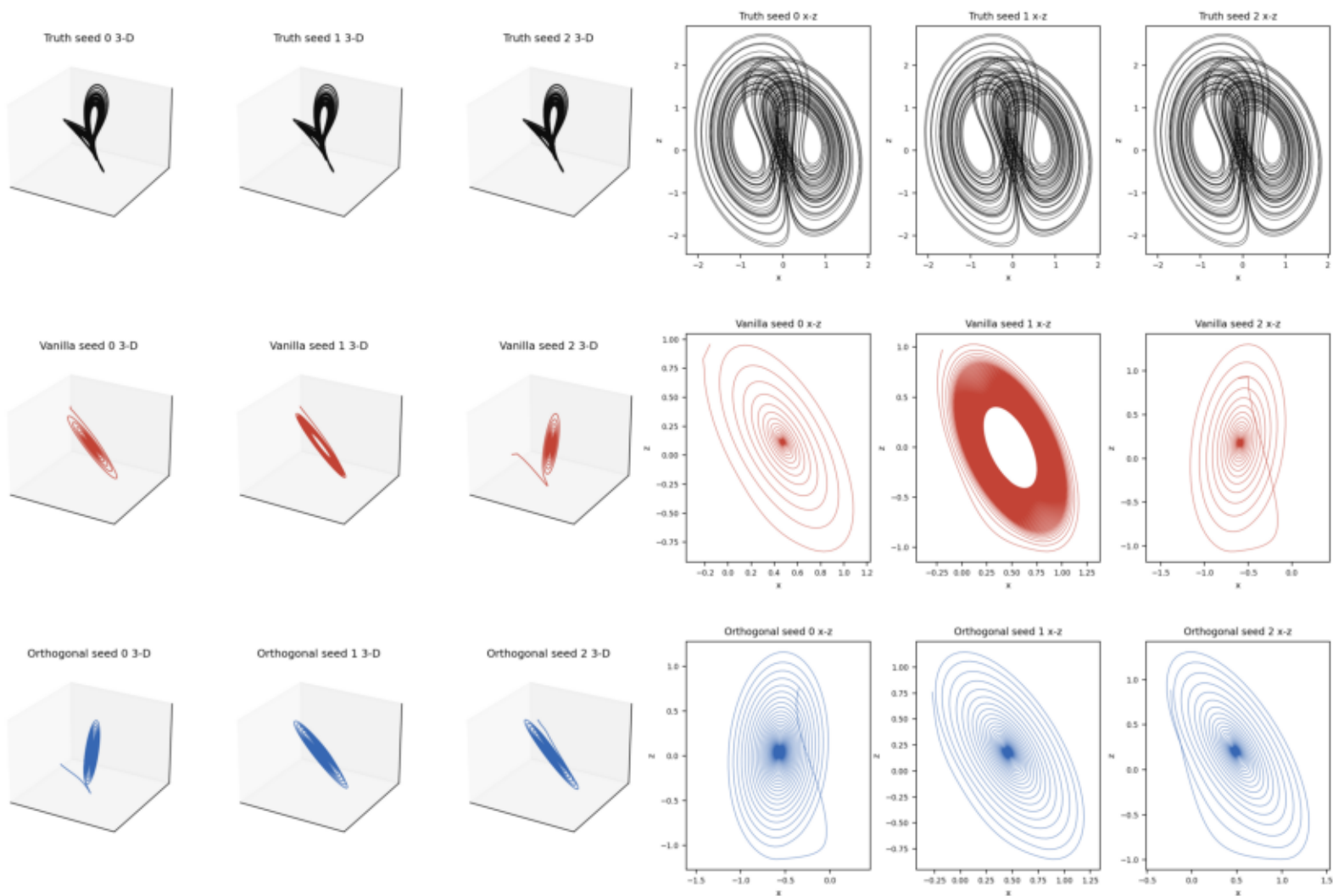
M = 4 P = 3 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.2987+/-0.1113 ortho 0.1582+/-0.0921 rel gap -47.0% D_H_state van 0.943+/-0.045 ortho 0.883+/-0.076 rel gap -6.4%



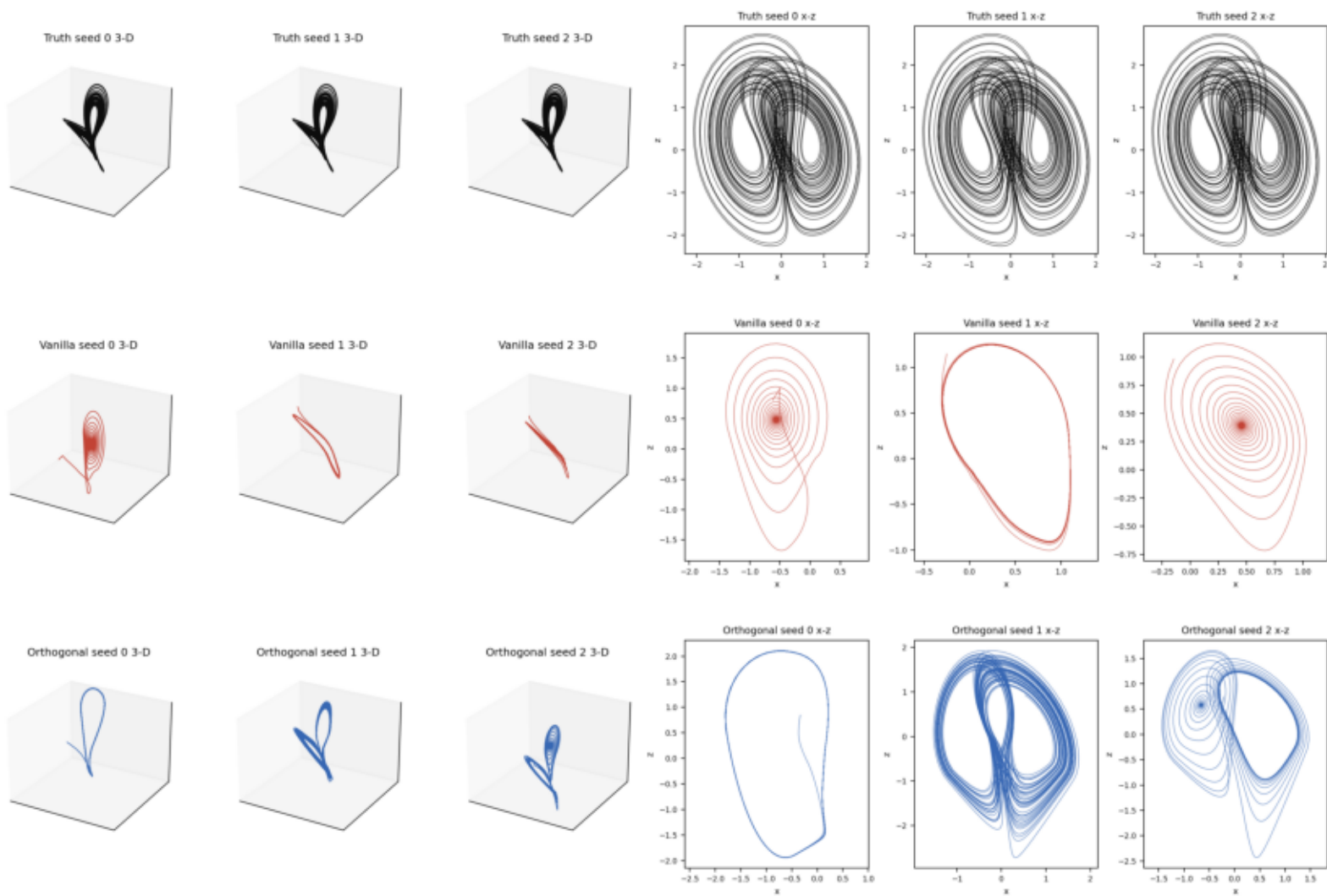
M = 4 P = 4 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3678+/-0.0158 ortho 0.4702+/-0.0992 rel gap +27.9% D_H_state van 0.930+/-0.013 ortho 0.916+/-0.059 rel gap -1.5%



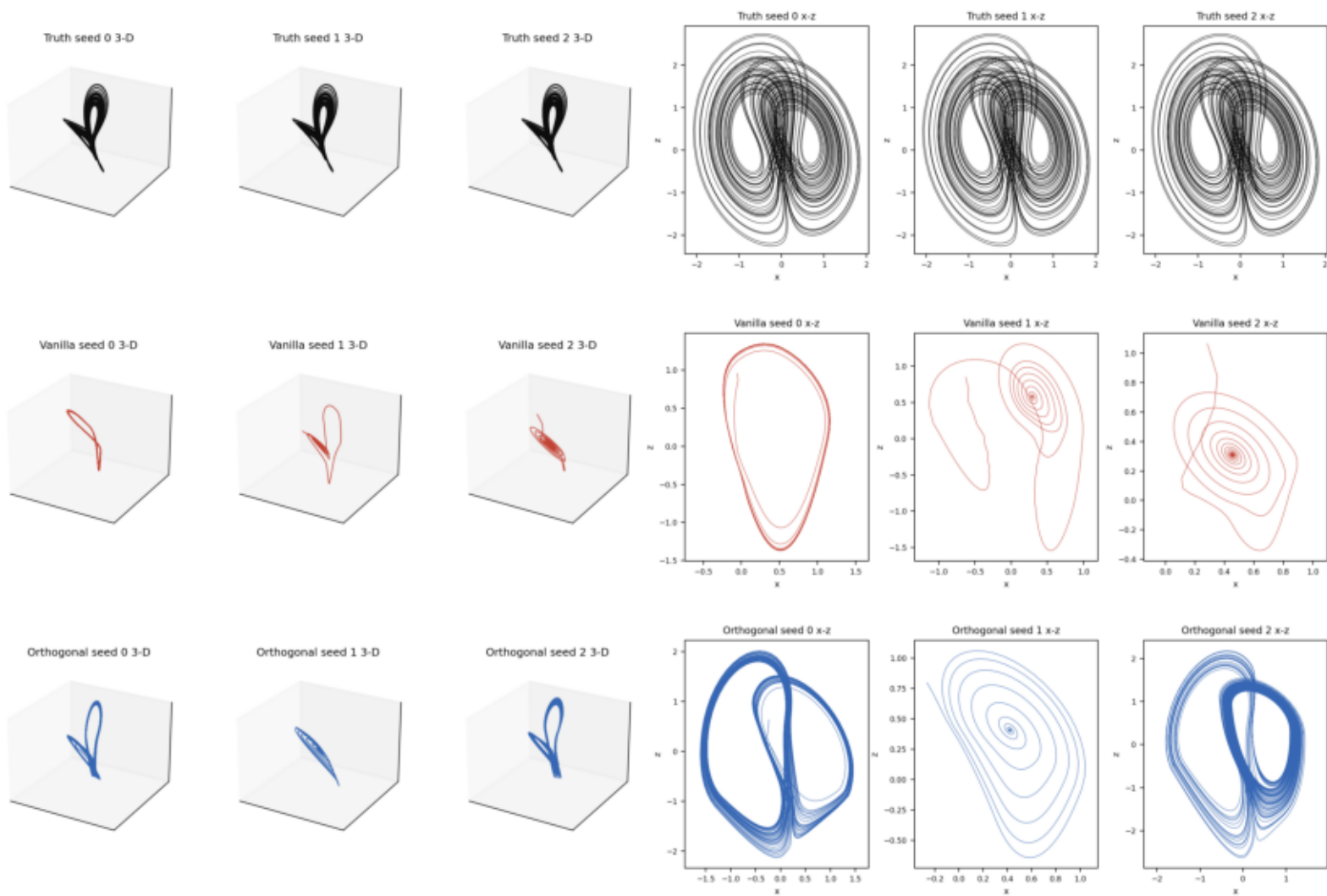
M = 5 P = 2 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.4409+/-0.0697 ortho 0.4170+/-0.0861 rel gap -5.4% D_H_state van 0.936+/-0.066 ortho 0.955+/-0.007 rel gap +2.0%



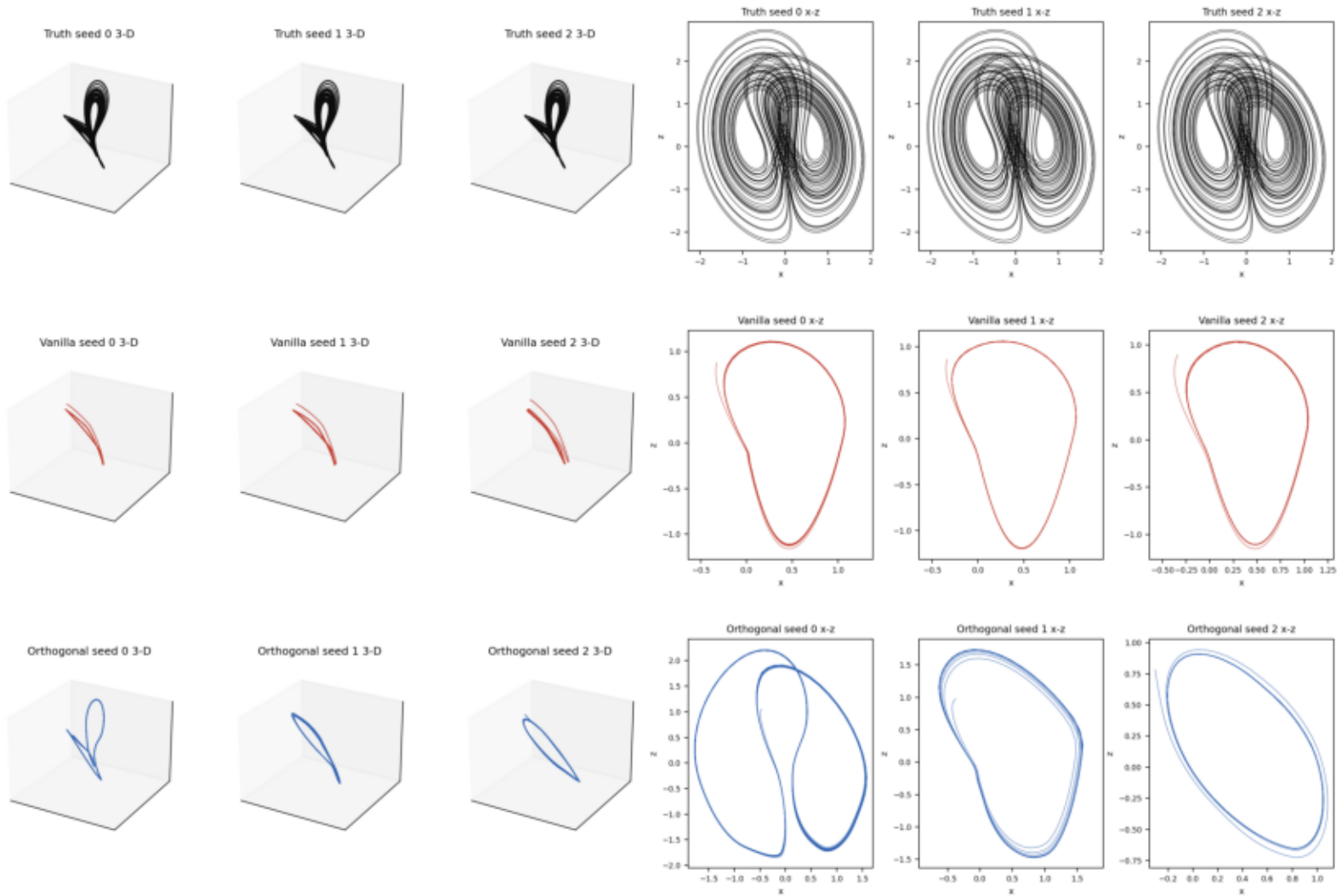
M = 5 P = 3 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.4091+/-0.1038 ortho 0.1653+/-0.0809 rel gap -59.6% D_H_state van 0.936+/-0.059 ortho 0.805+/-0.136 rel gap -14.0%



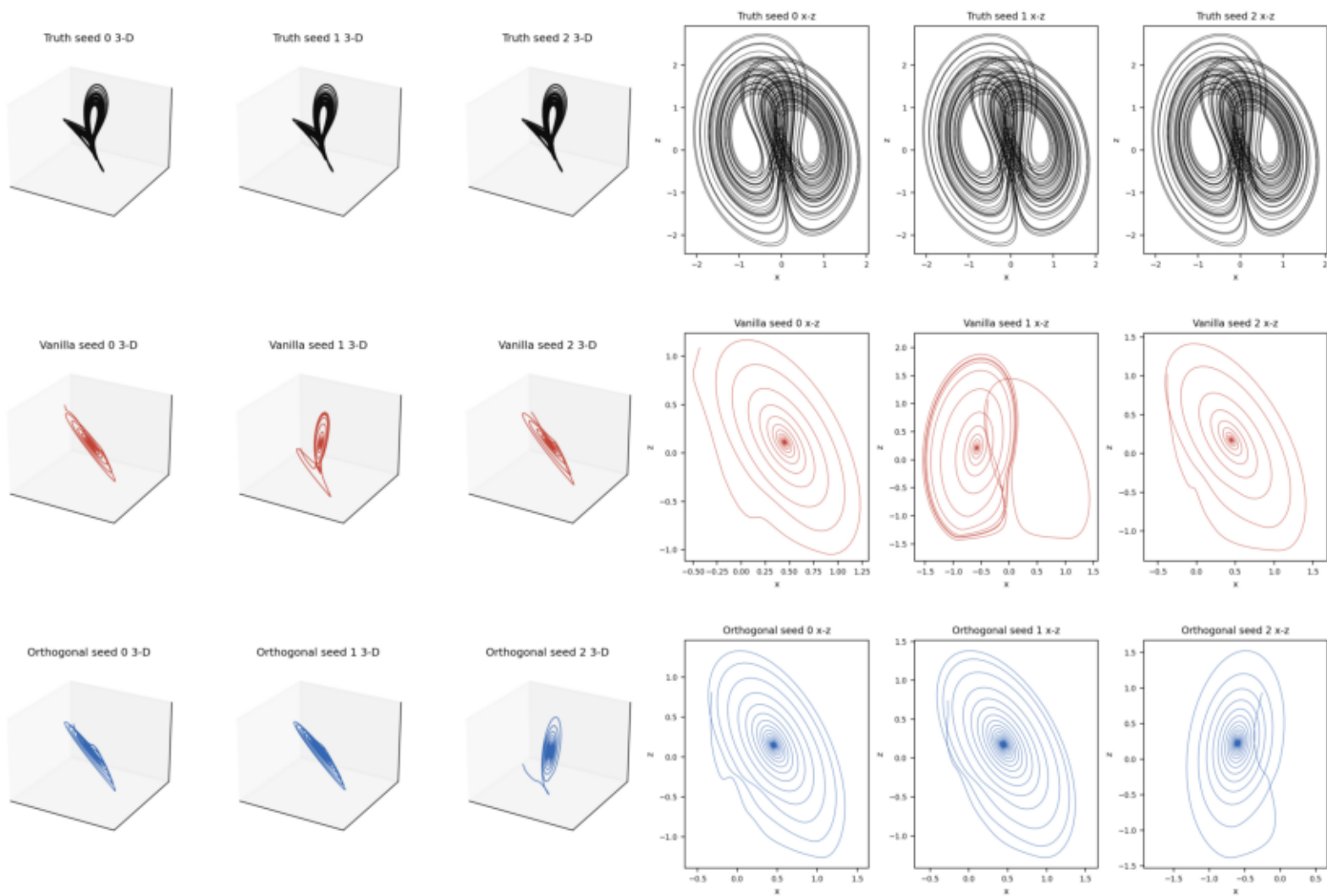
M = 5 P = 4 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3059+/-0.1535 ortho 0.1800+/-0.1720 rel gap -41.1% D_H_state van 0.968+/-0.036 ortho 0.798+/-0.156 rel gap -17.6%



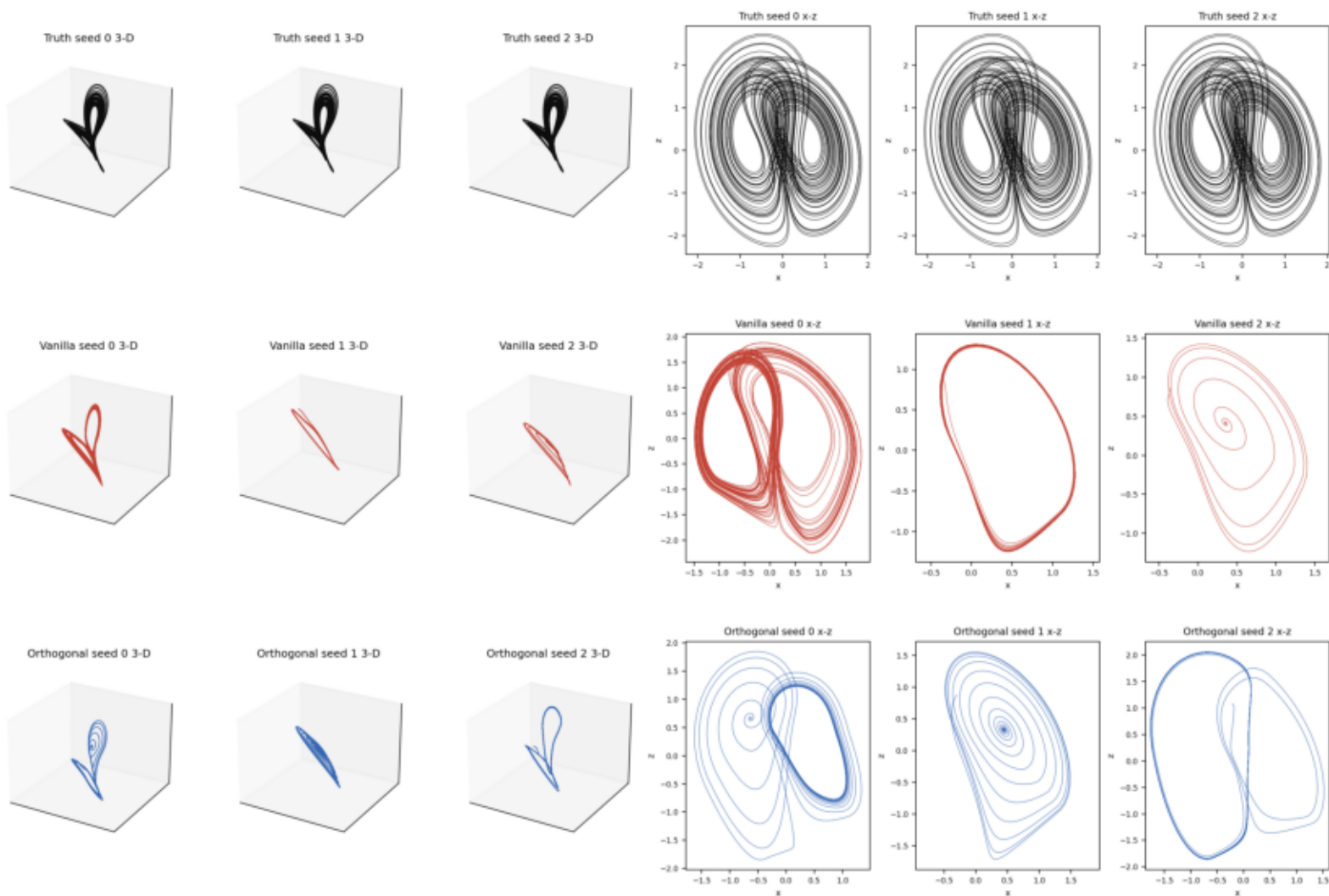
M = 5 P = 5 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3889+/-0.0062 ortho 0.3101+/-0.1982 rel gap -20.3% D_H_state van 0.907+/-0.037 ortho 0.884+/-0.011 rel gap -2.5%



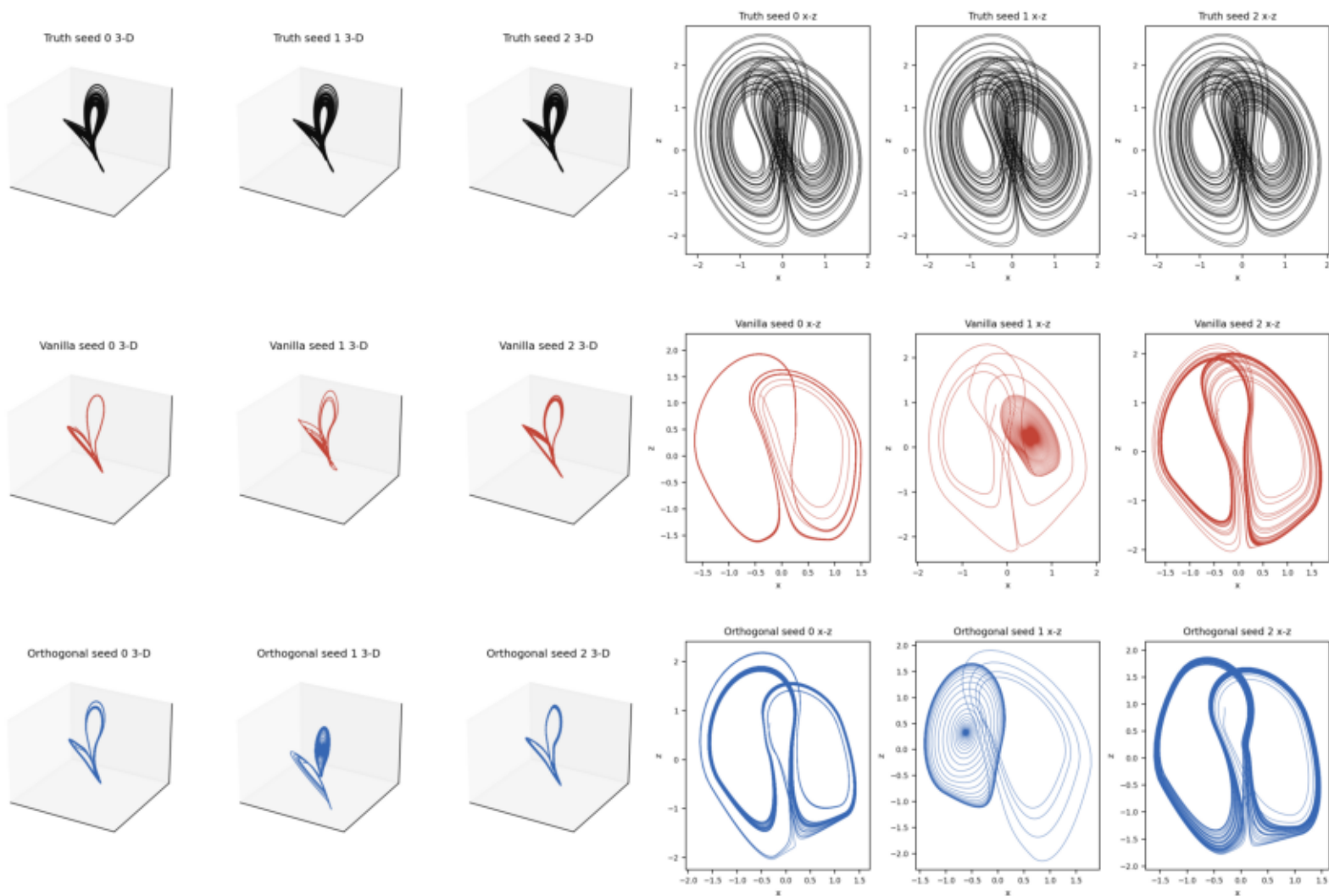
M = 8 P = 2 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.2548+/-0.1630 ortho 0.3339+/-0.0412 rel gap +31.1% D_H_state van 0.950+/-0.022 ortho 0.957+/-0.016 rel gap +0.7%



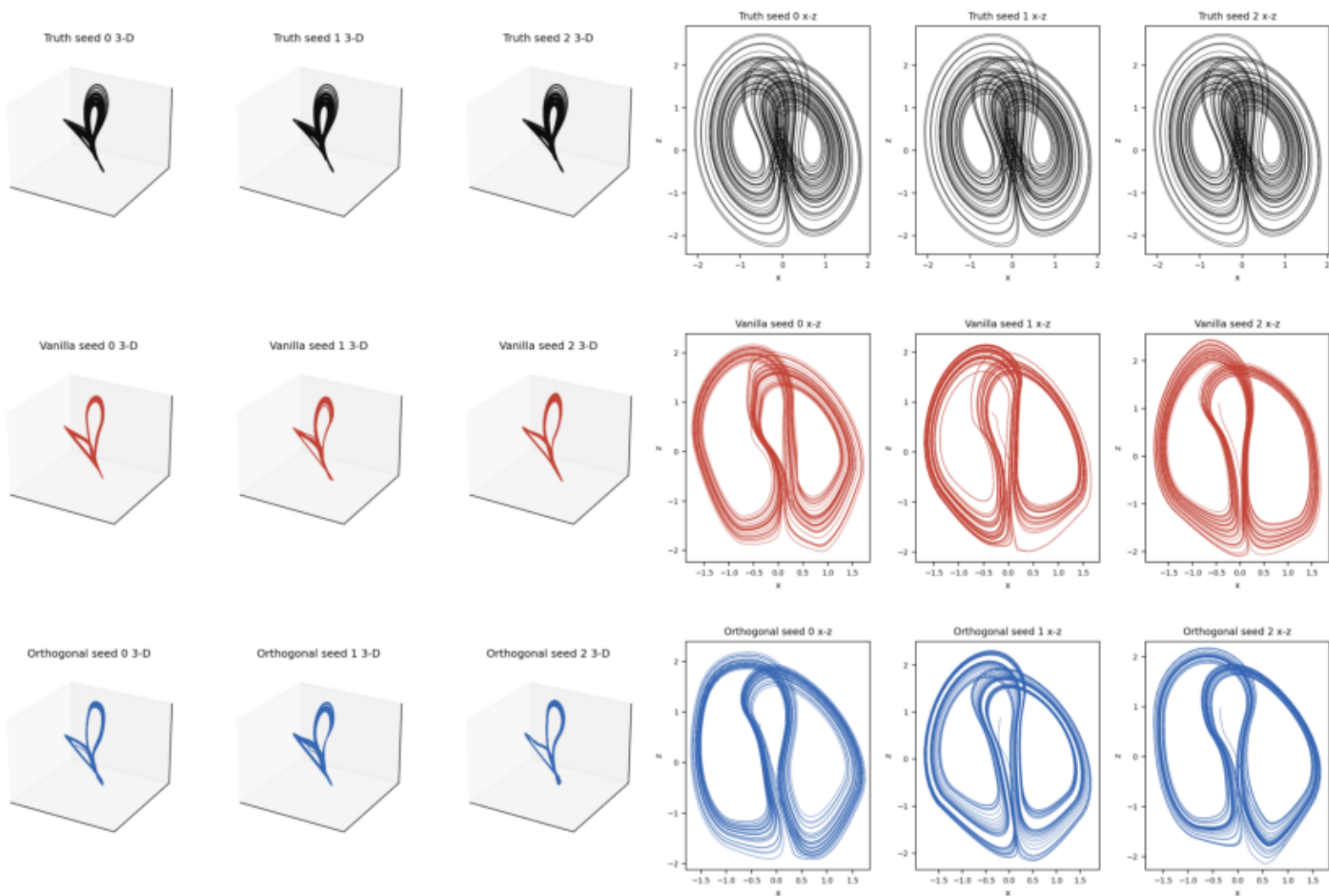
M = 8 P = 3 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.2438+/-0.1650 ortho 0.2120+/-0.0207 rel gap -13.1% D_H_state van 0.845+/-0.124 ortho 0.888+/-0.023 rel gap +5.0%



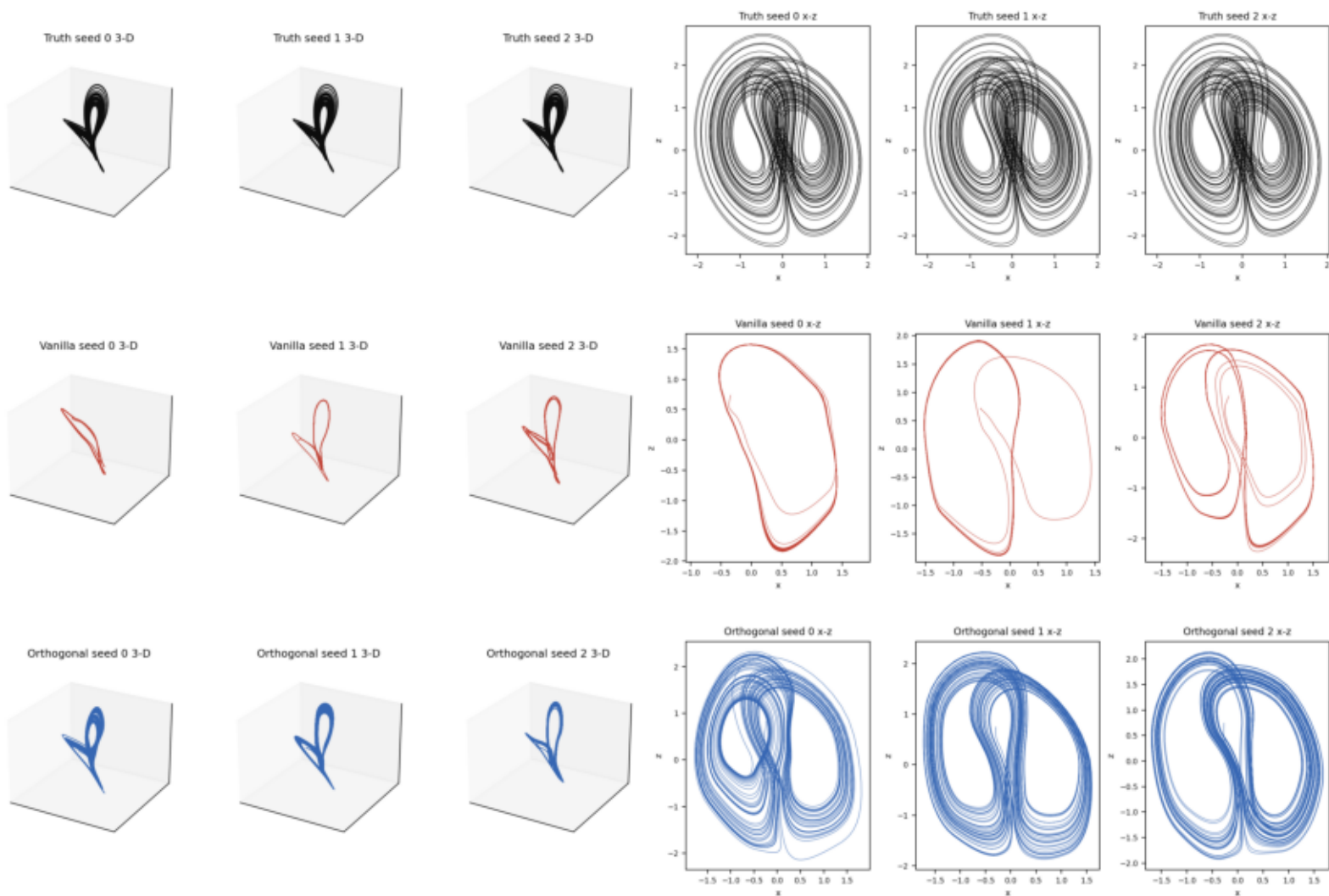
M = 8 P = 4 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.1158+/-0.0318 ortho 0.0833+/-0.0053 rel gap -28.1% D_H_state van 0.786+/-0.086 ortho 0.761+/-0.088 rel gap -3.1%



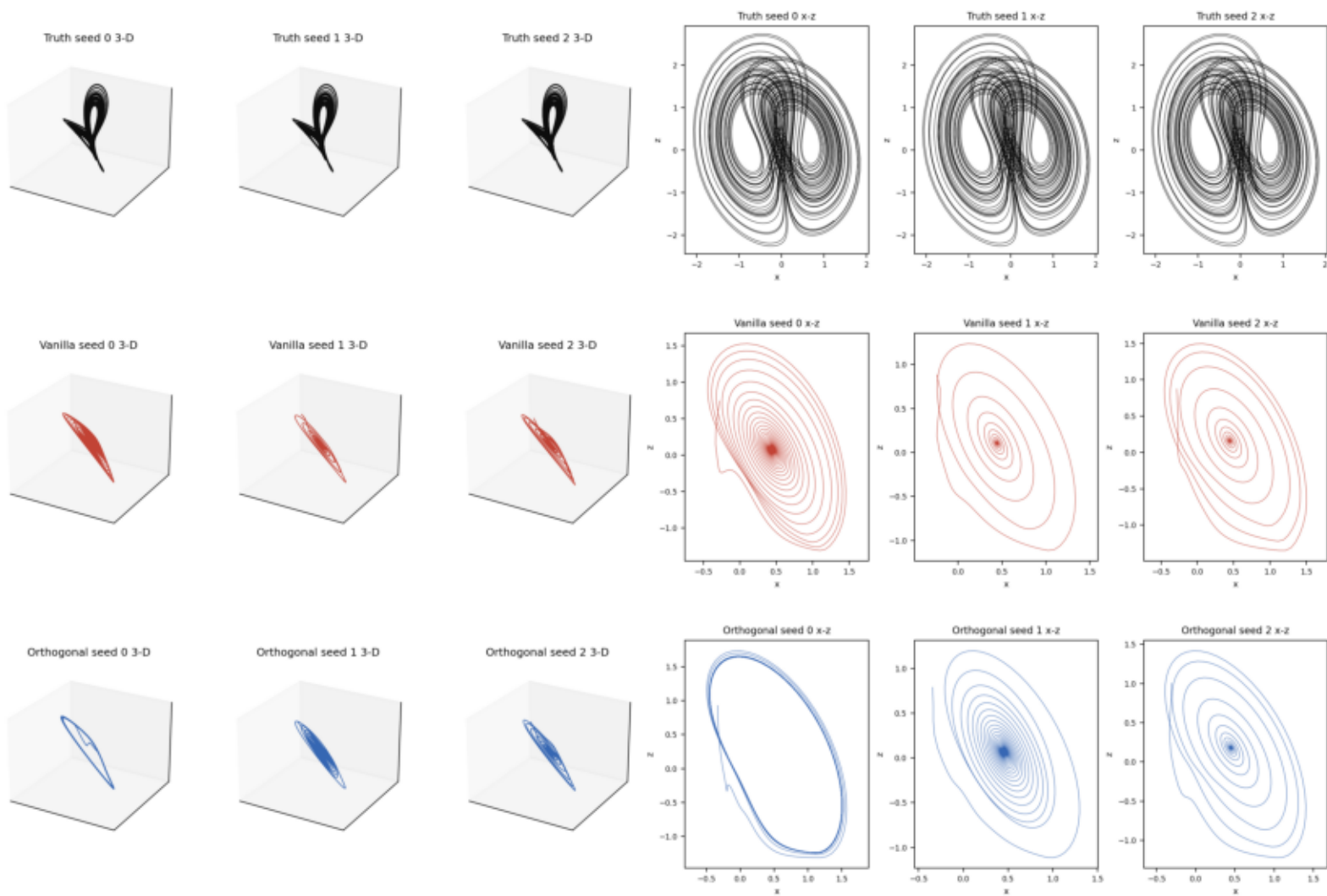
M = 8 P = 5 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.0800+/-0.0064 ortho 0.0864+/-0.0040 rel gap +8.0% D_H_state van 0.646+/-0.039 ortho 0.645+/-0.038 rel gap -0.1%



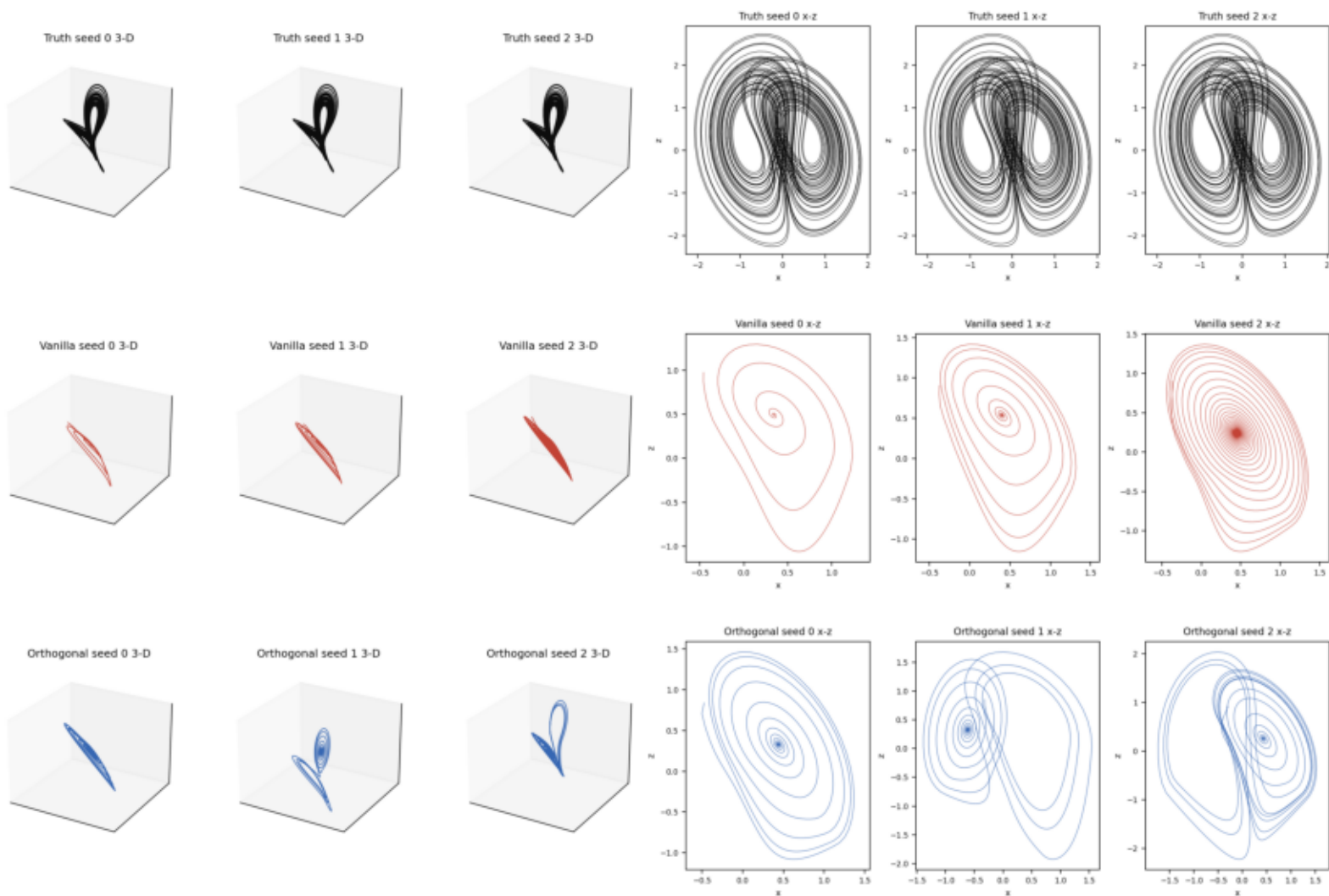
M = 8 P = 6 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.1810+/-0.0889 ortho 0.0688+/-0.0217 rel gap -62.0% D_H_state van 0.834+/-0.045 ortho 0.609+/-0.059 rel gap -27.0%



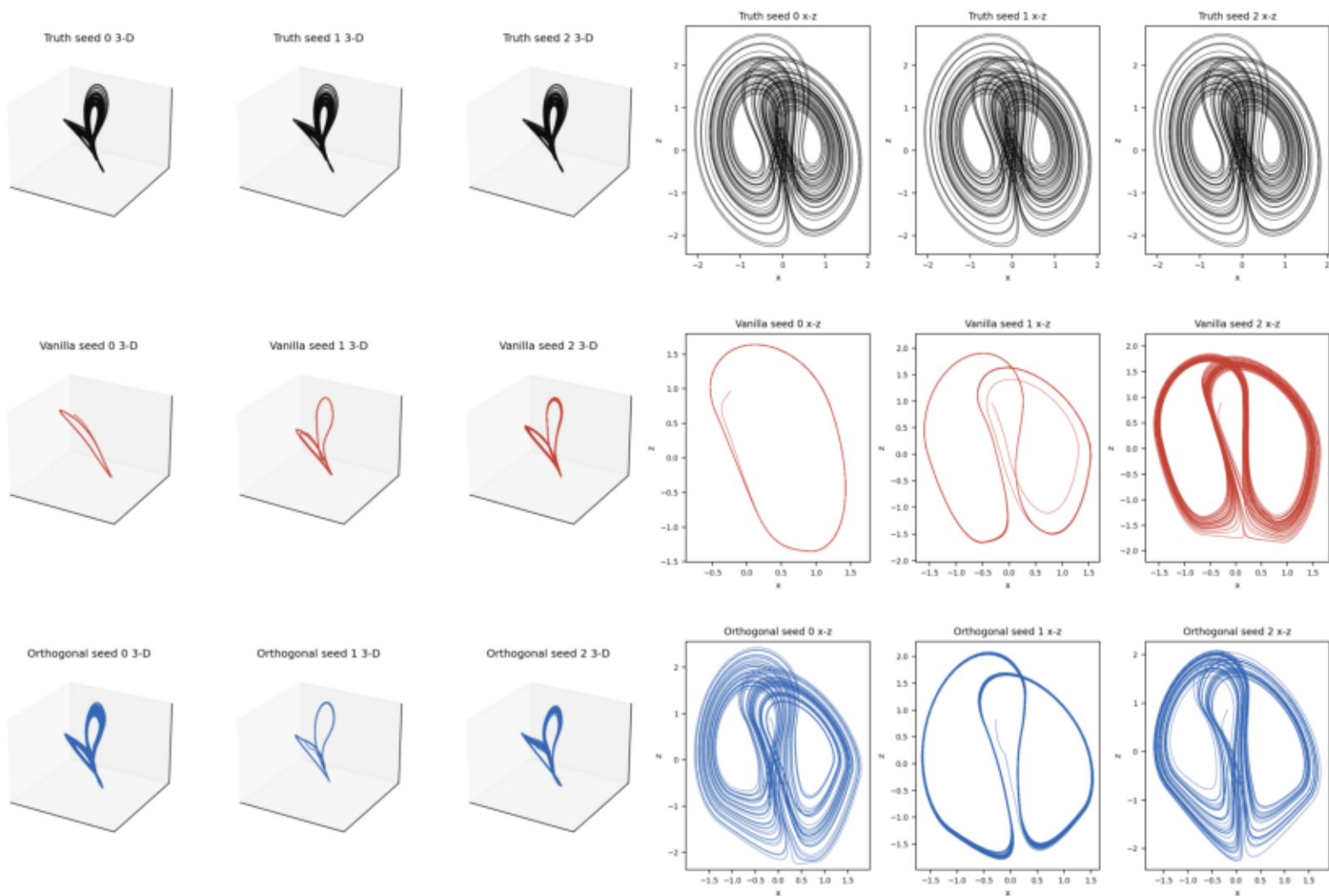
M = 9 P = 2 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3532 $\pm$ 0.0495 ortho 0.3653 $\pm$ 0.0322 rel gap +3.4% D_H_state van 0.943 $\pm$ 0.028 ortho 0.919 $\pm$ 0.055 rel gap -2.5%



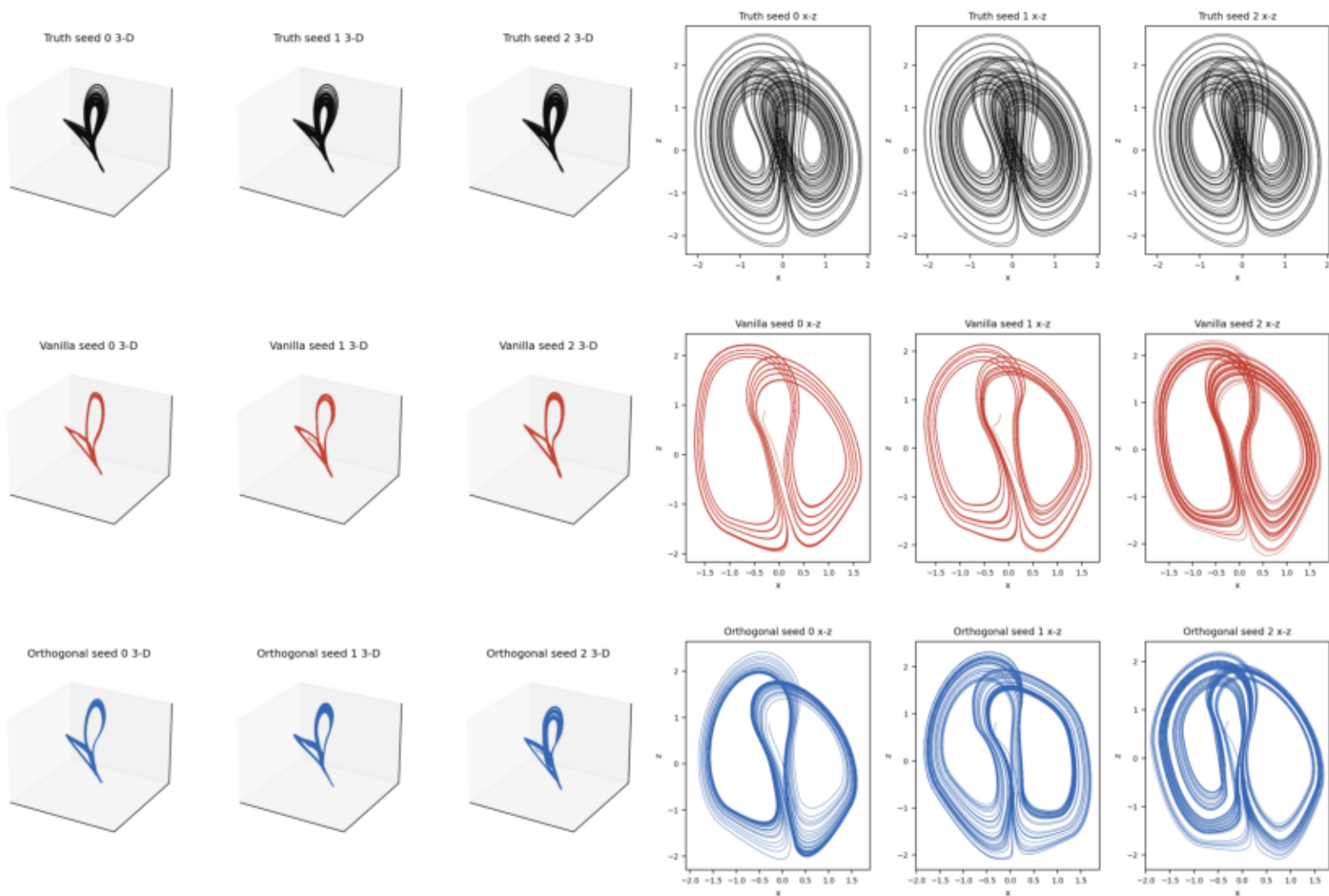
M = 9 P = 3 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.3270+/-0.0147 ortho 0.1855+/-0.1261 rel gap -43.3% D_H_state van 0.945+/-0.024 ortho 0.921+/-0.034 rel gap -2.5%



M = 9 P = 4 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.1846+/-0.1591 ortho 0.0699+/-0.0273 rel gap -62.1% D_H_state van 0.801+/-0.076 ortho 0.625+/-0.149 rel gap -22.0%



M = 9 P = 5 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.0867+/-0.0015 ortho 0.0770+/-0.0113 rel gap -11.1% D_H_state van 0.650+/-0.006 ortho 0.635+/-0.024 rel gap -2.3%



M = 9 P = 6 (all 3 seeds; cols 1-3: 3-D, cols 4-6: x-z 2-D)
PSE van 0.0718+/-0.0069 ortho 0.0724+/-0.0075 rel gap +0.9% D_H_state van 0.583+/-0.038 ortho 0.615+/-0.054 rel gap +5.5%

